

St. Martins Marsh Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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Funding & Acknowledgements

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

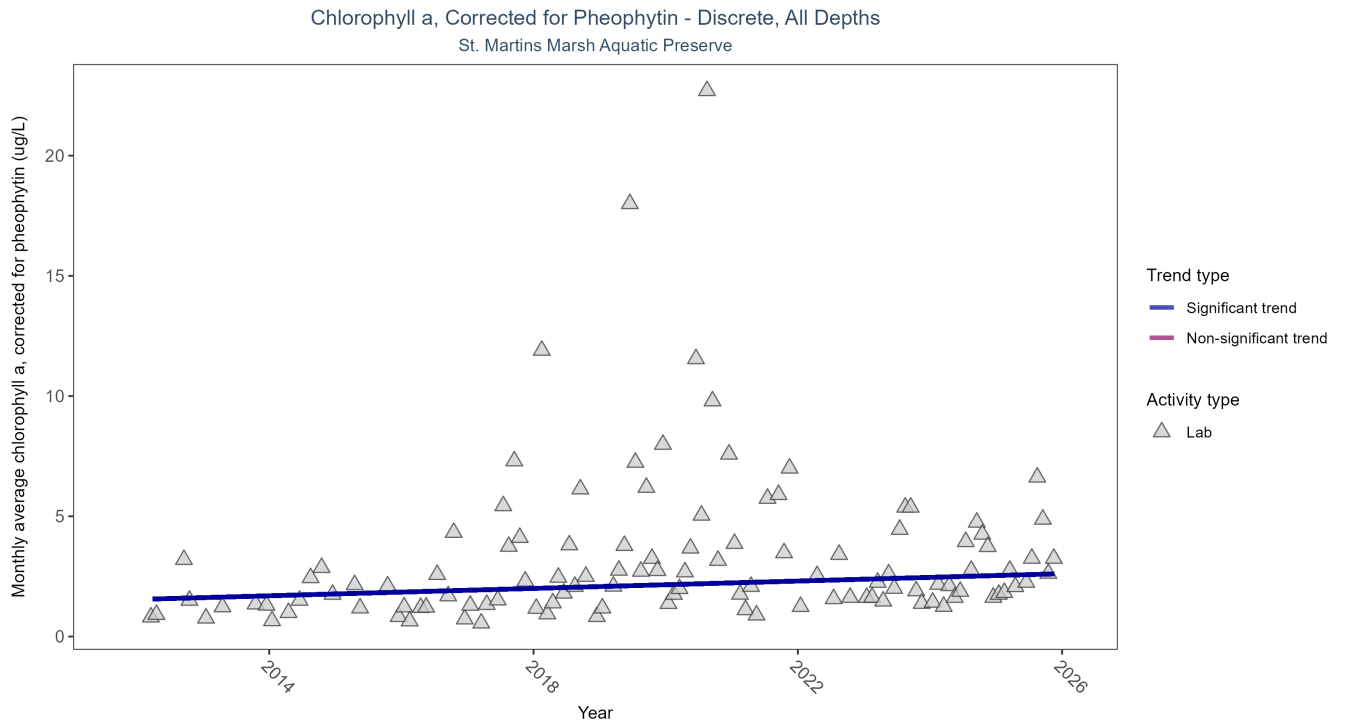


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	635	14	2012 - 2025	1.8	0.2774	1.5416	0.0766	0.0001

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.08 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

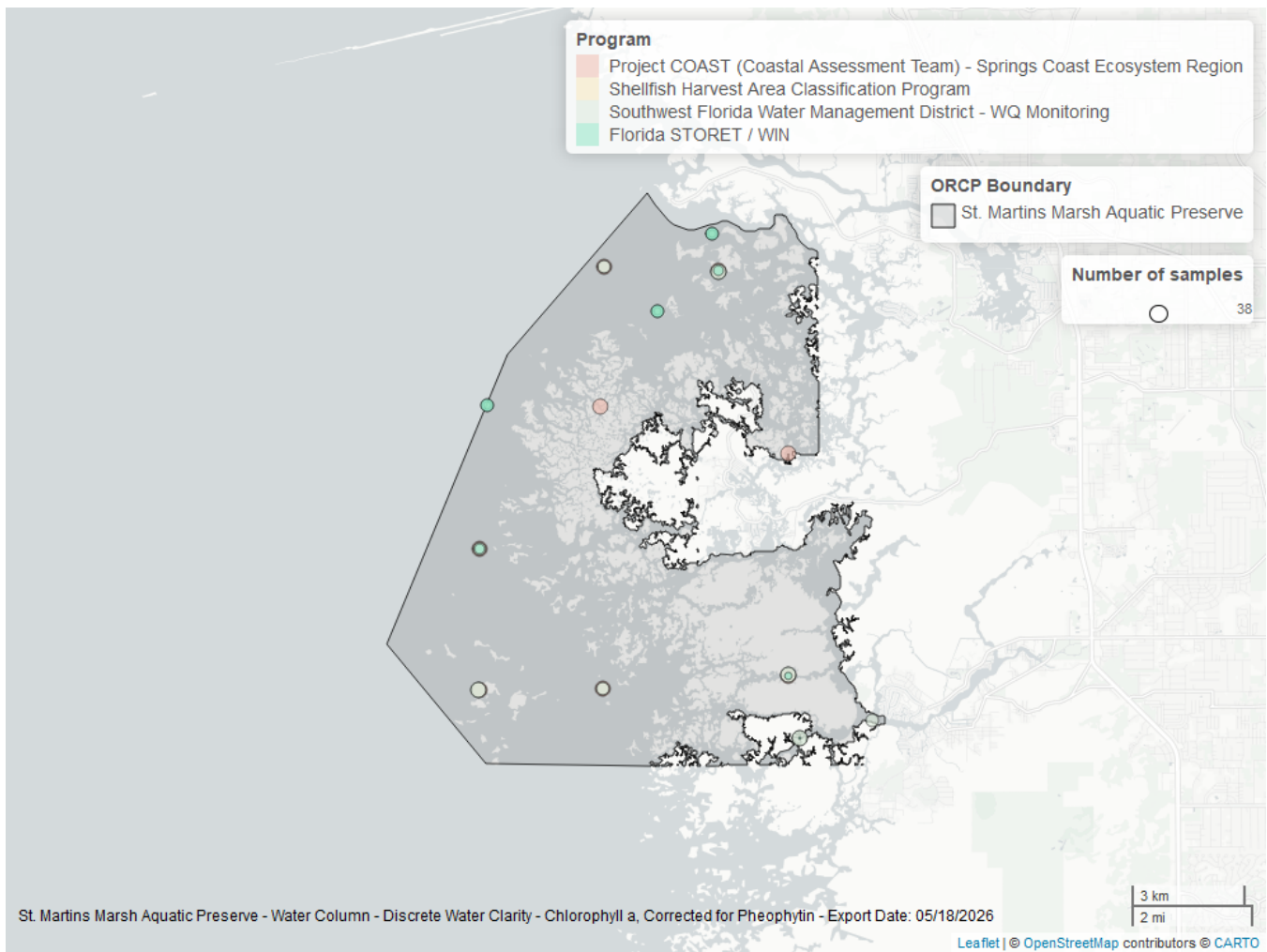


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5008	256	2023	2025
479	247	2016	2025
5002	116	2012	2021
540	21	2017	2019

Program names:

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region¹

479 - Southwest Florida Water Management District - Water Quality Monitoring²

5002 - Florida STORET / WIN³

540 - Shellfish Harvest Area Classification Program⁴

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

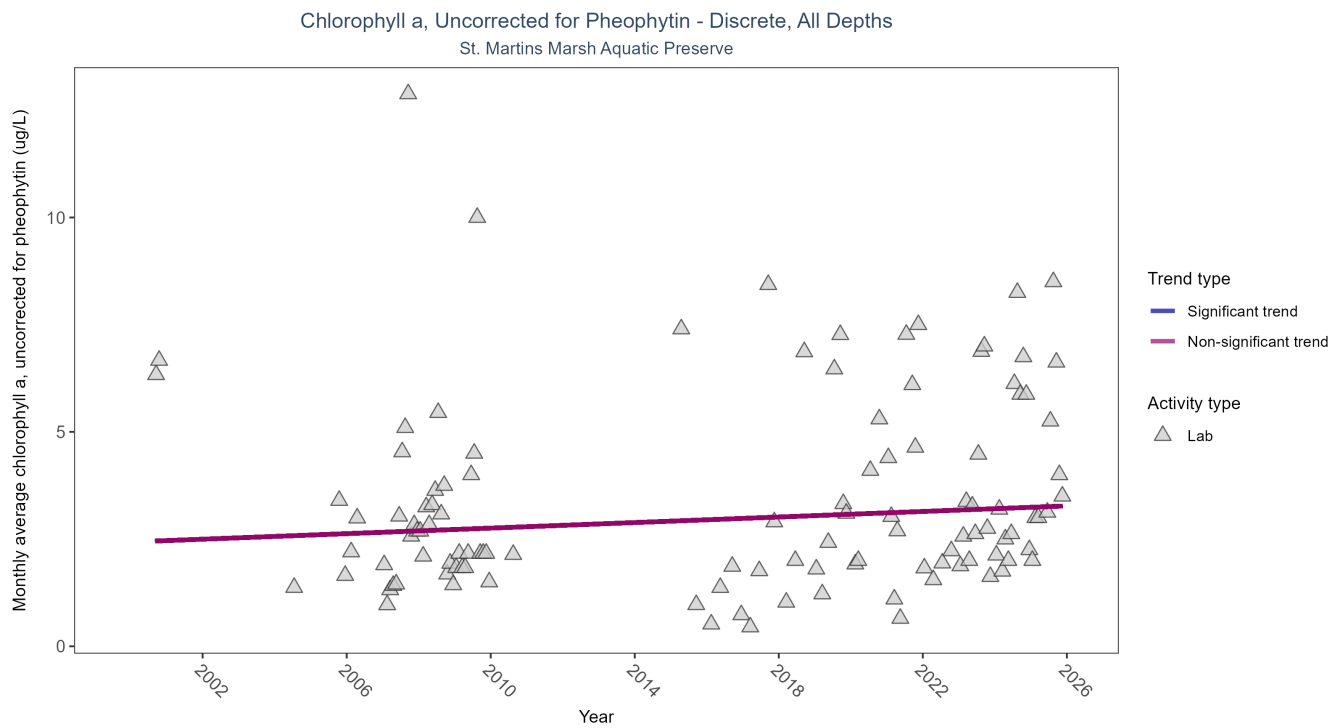


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	604	19	2000 - 2025	2.05	0.1176	2.4324	0.0323	0.1048

Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 2000 and 2025.

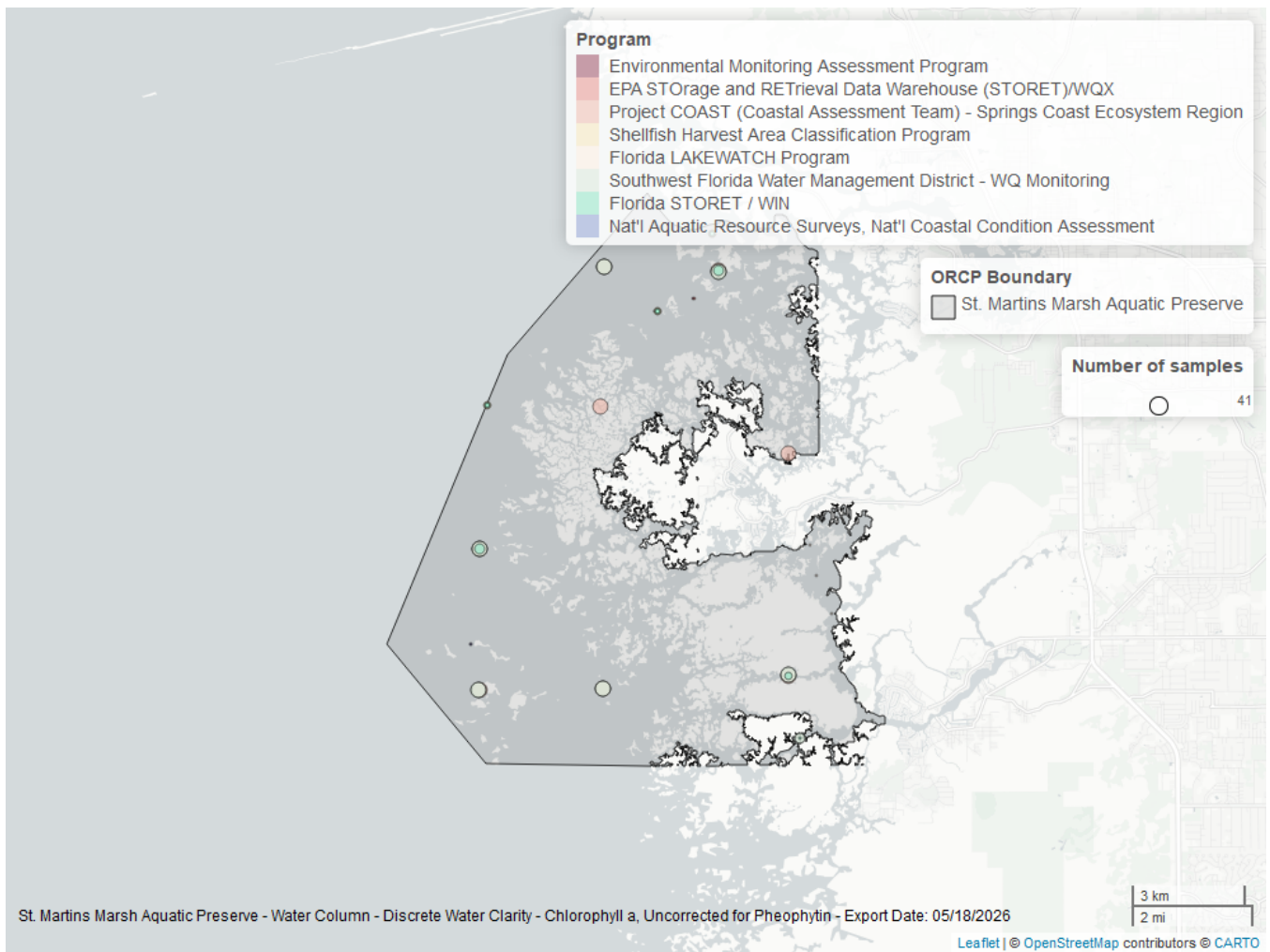


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5008	260	2022	2025
479	259	2005	2024
5002	60	2015	2021
540	21	2017	2019
514	6	2000	2000
103	2	2004	2015
118	1	2010	2010
115	1	2004	2004

Program names:

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region¹

479 - Southwest Florida Water Management District - Water Quality Monitoring²

5002 - Florida STORET / WIN³

540 - Shellfish Harvest Area Classification Program⁴

514 - Florida LAKEWATCH Program⁵
 103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX⁶
 118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁷
 115 - Environmental Monitoring Assessment Program⁸

Colored Dissolved Organic Matter - Discrete

Seasonal Kendall-Tau Trend Analysis

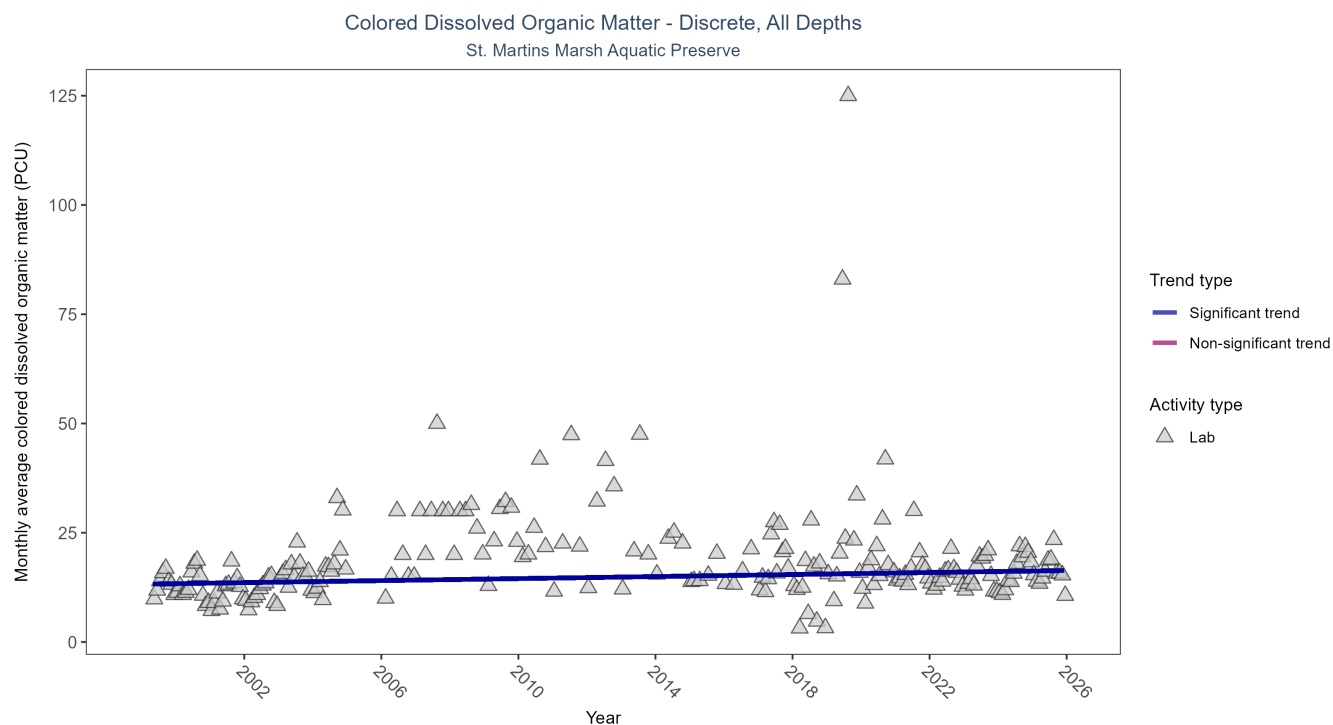


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1203	26	1999 - 2025	13	0.1644	13.2338	0.1157	0.0014

Monthly average colored dissolved organic matter increased by 0.12 PCU per year, indicating a decrease in water clarity.

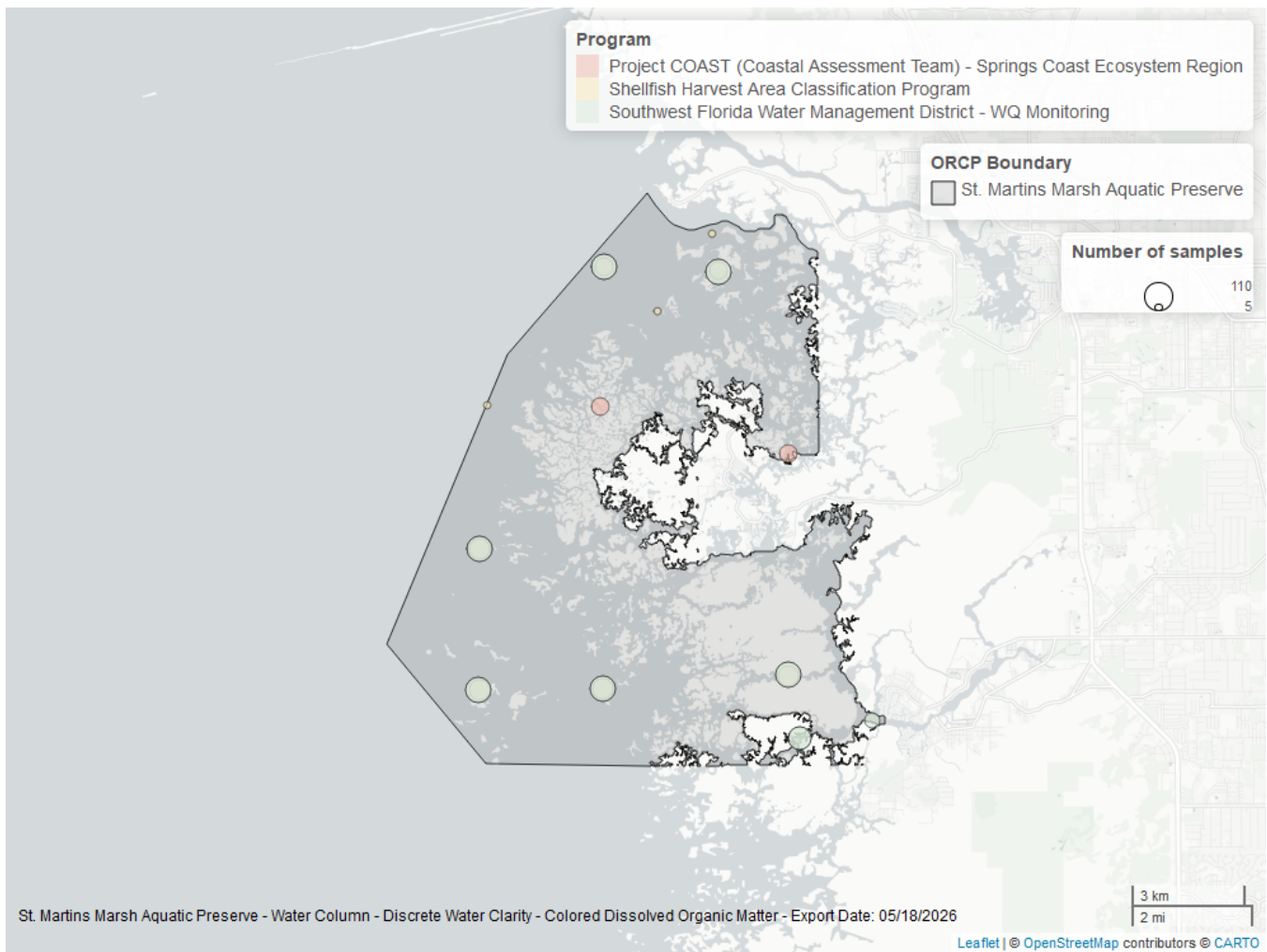


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

ProgramID	N_Data	YearMin	YearMax
479	738	1999	2025
5008	448	2021	2025
540	21	2017	2019

Program names:

479 - Southwest Florida Water Management District - Water Quality Monitoring²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region¹

540 - Shellfish Harvest Area Classification Program⁴

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

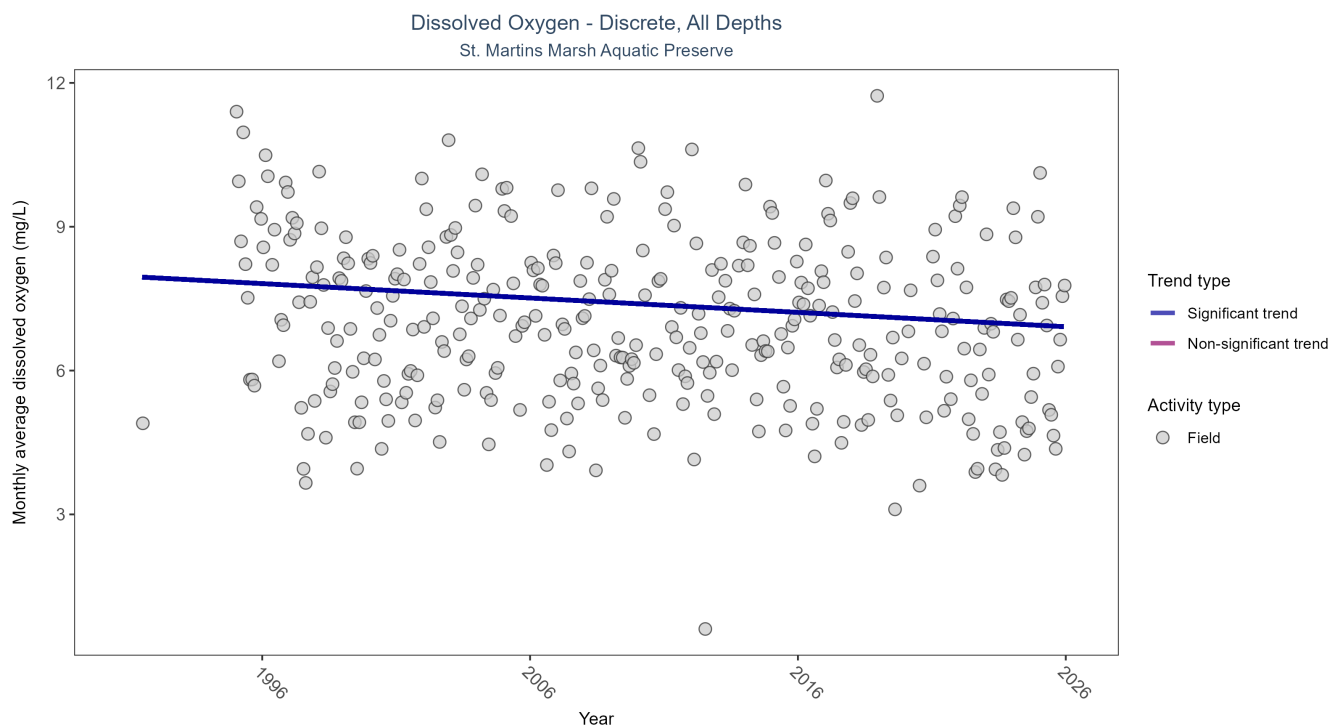


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	8732	32	1991 - 2025	7.25	-0.1606	7.9637	-0.03	0

Monthly average dissolved oxygen decreased by 0.03 mg/L per year.

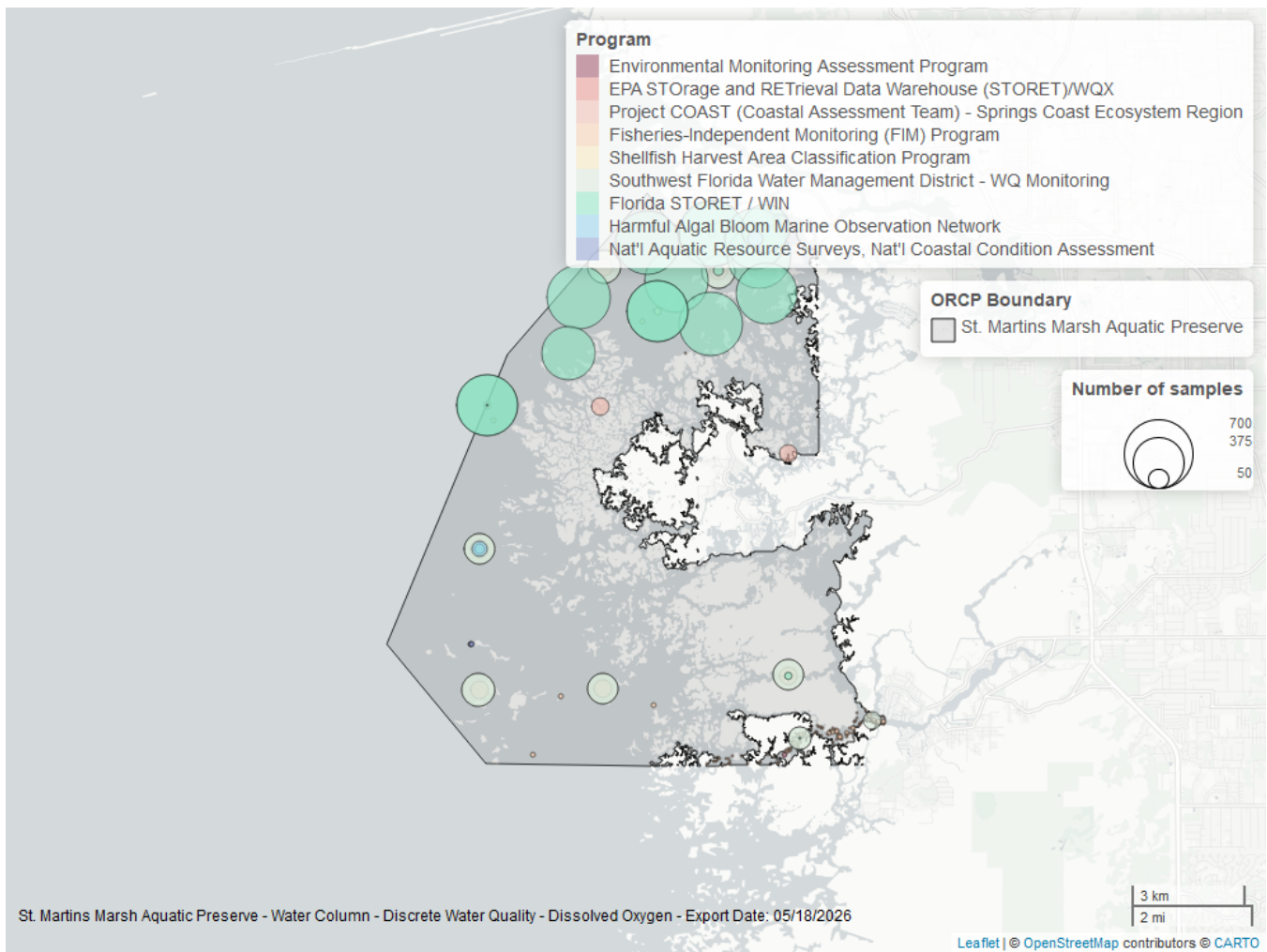


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	6897	1995	2021
479	1072	1996	2025
5008	448	2021	2025
69	220	2003	2008
95	72	2011	2018
540	21	2017	2019
118	9	2015	2020
115	5	1991	2004
103	3	2015	2015

Program names:

5002 - Florida STORET / WIN³

479 - Southwest Florida Water Management District - Water Quality Monitoring²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region¹

69 - Fisheries-Independent Monitoring (FIM) Program⁹
95 - Harmful Algal Bloom Marine Observation Network¹⁰
540 - Shellfish Harvest Area Classification Program⁴
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁷
115 - Environmental Monitoring Assessment Program⁸
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

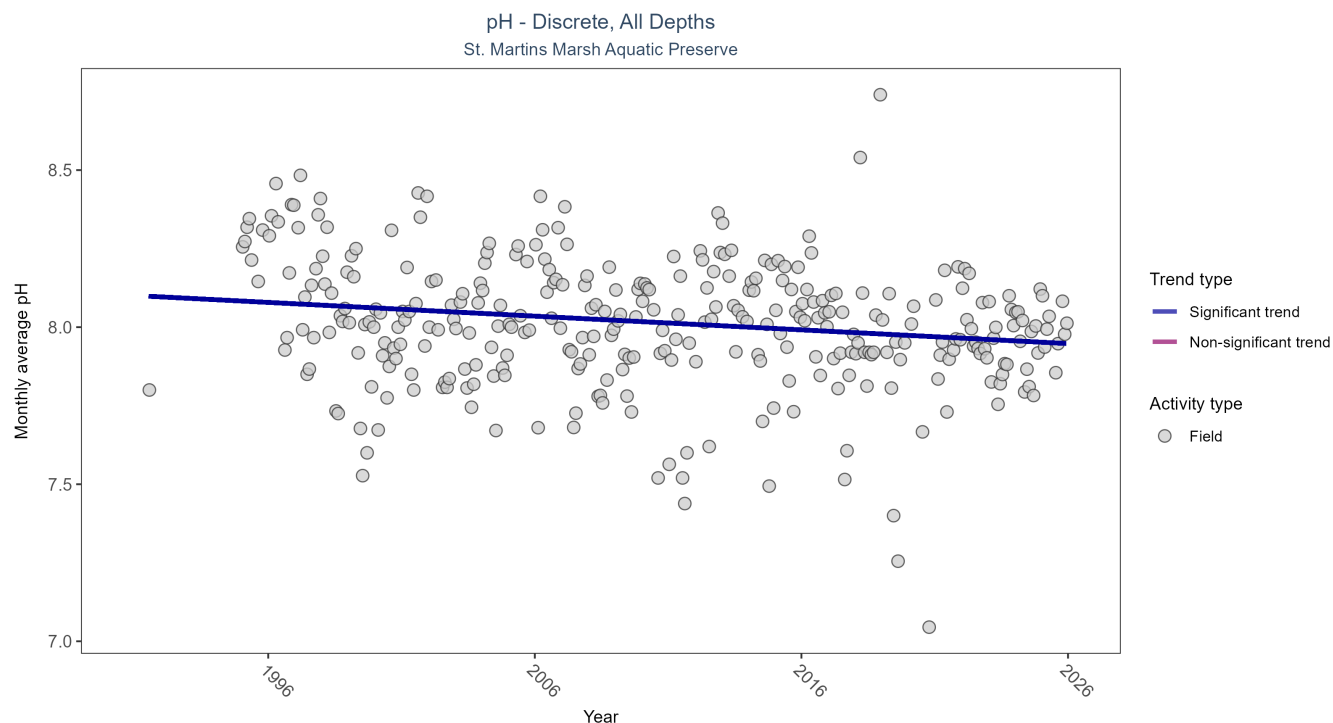


Figure 9: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	4817	32	1991 - 2025	8.06	-0.1512	8.1007	-0.0044	0.0001

Monthly average pH decreased by less than 0.01 pH units per year.

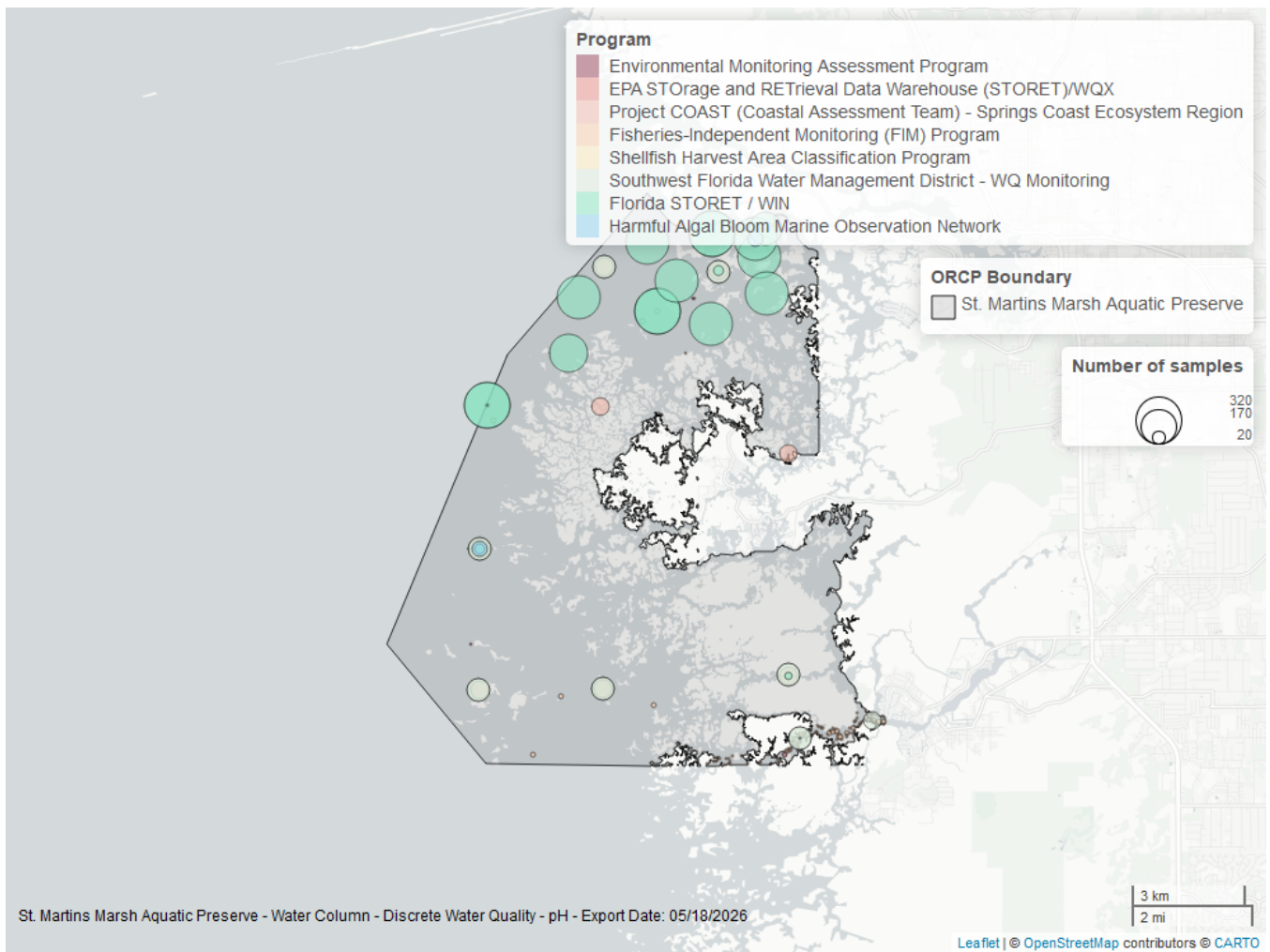


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5002	3487	1995	2021
479	621	2003	2025
5008	423	2021	2025
69	220	2003	2008
95	70	2011	2018
540	9	2017	2018
115	5	1991	2004
103	2	2015	2015

Program names:

5002 - Florida STORET / WIN³

479 - Southwest Florida Water Management District - Water Quality Monitoring²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region¹

69 - Fisheries-Independent Monitoring (FIM) Program⁹

95 - Harmful Algal Bloom Marine Observation Network¹⁰
540 - Shellfish Harvest Area Classification Program⁴
115 - Environmental Monitoring Assessment Program⁸
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

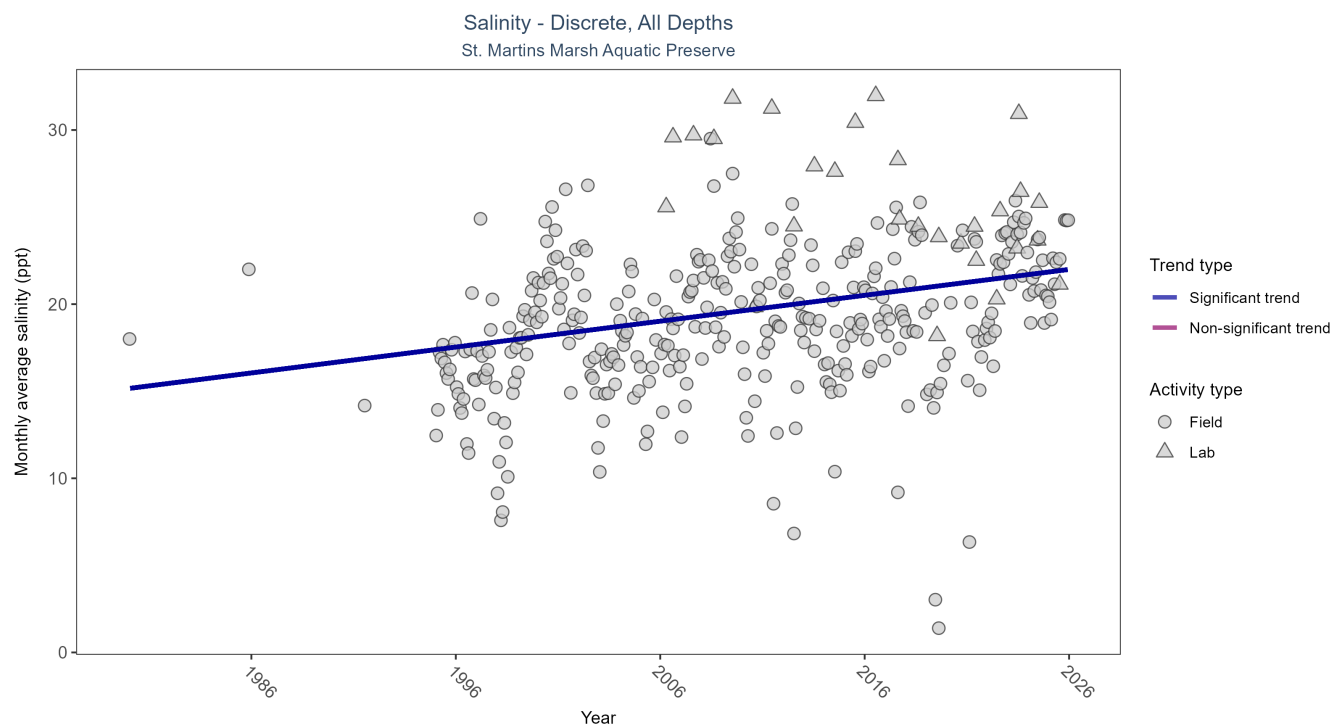


Figure 11: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	9658	34	1980 - 2025	18.4	0.2592	15.158	0.1486	0

Monthly average salinity increased by 0.15 ppt per year.

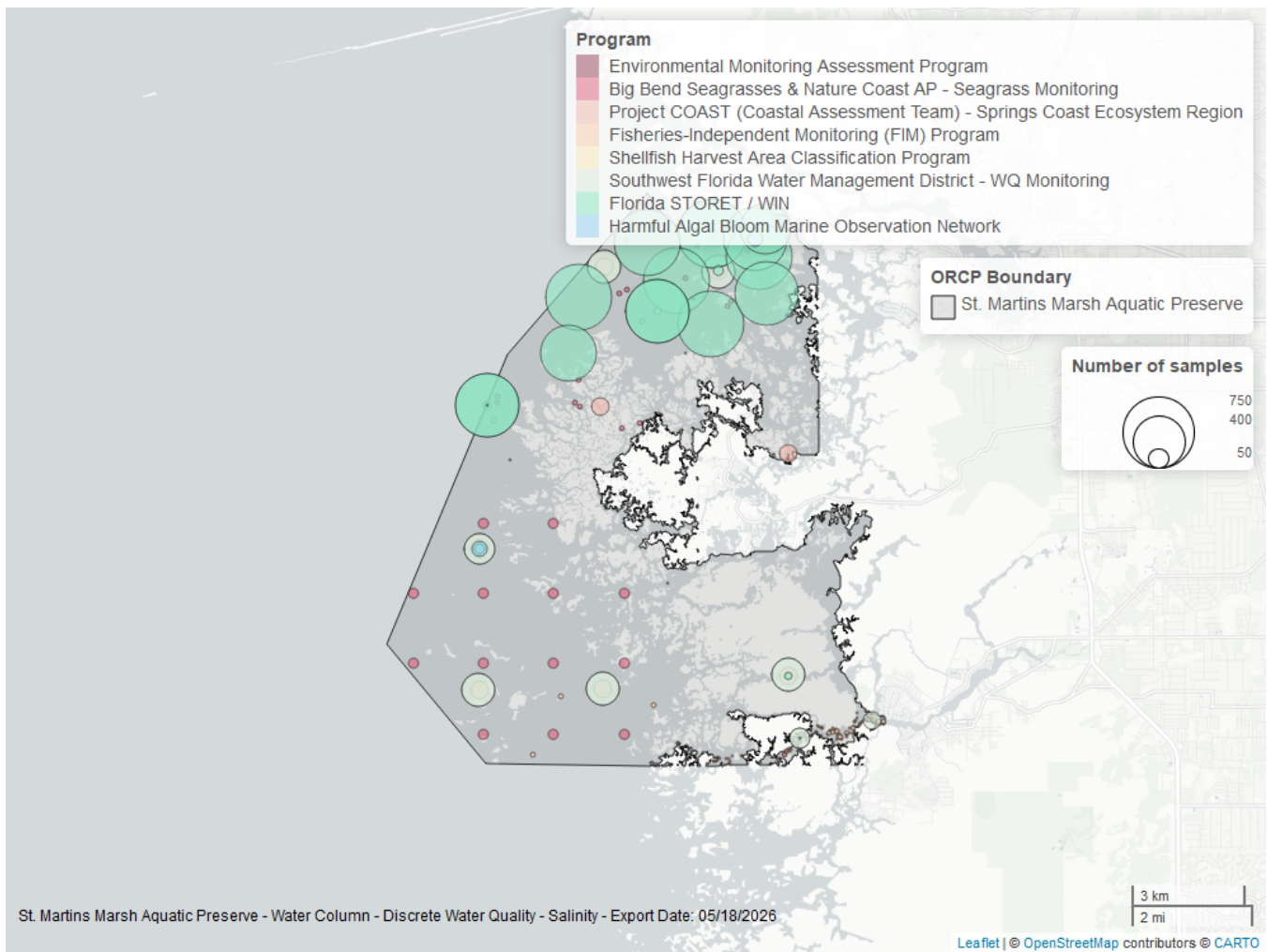


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5002	7536	1995	2021
479	1066	1996	2025
5008	437	2021	2025
560	326	2006	2025
69	220	2003	2008
95	77	1980	2018
540	21	2017	2019
115	5	1991	2004

Program names:

5002 - Florida STORET / WIN³

479 - Southwest Florida Water Management District - Water Quality Monitoring²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region¹

560 - Big Bend Seagrasses & Nature Coast Aquatic Preserves - Seagrass Monitoring¹¹

69 - Fisheries-Independent Monitoring (FIM) Program⁹
95 - Harmful Algal Bloom Marine Observation Network¹⁰
540 - Shellfish Harvest Area Classification Program⁴
115 - Environmental Monitoring Assessment Program⁸

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

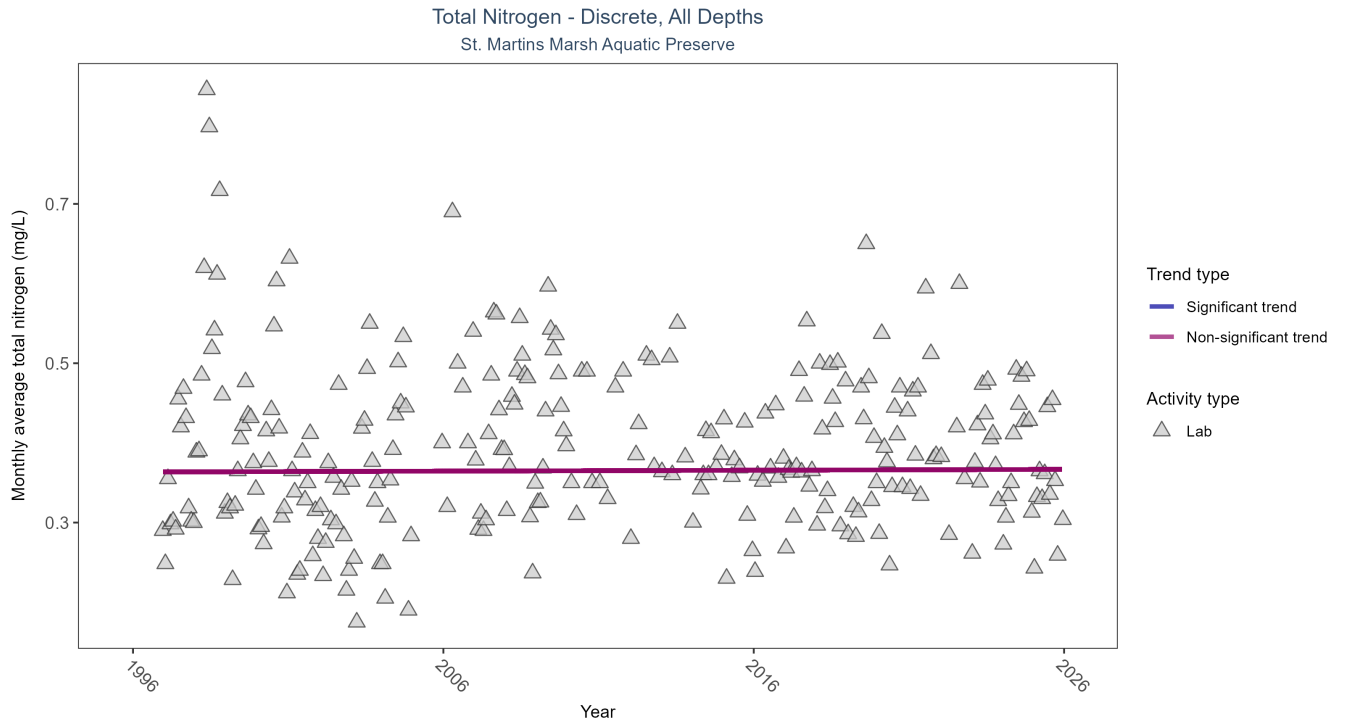


Figure 13: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1537	30	1996 - 2025	0.36	0.0247	0.3635	0.0001	0.8288

Total nitrogen showed no detectable trend between 1996 and 2025.

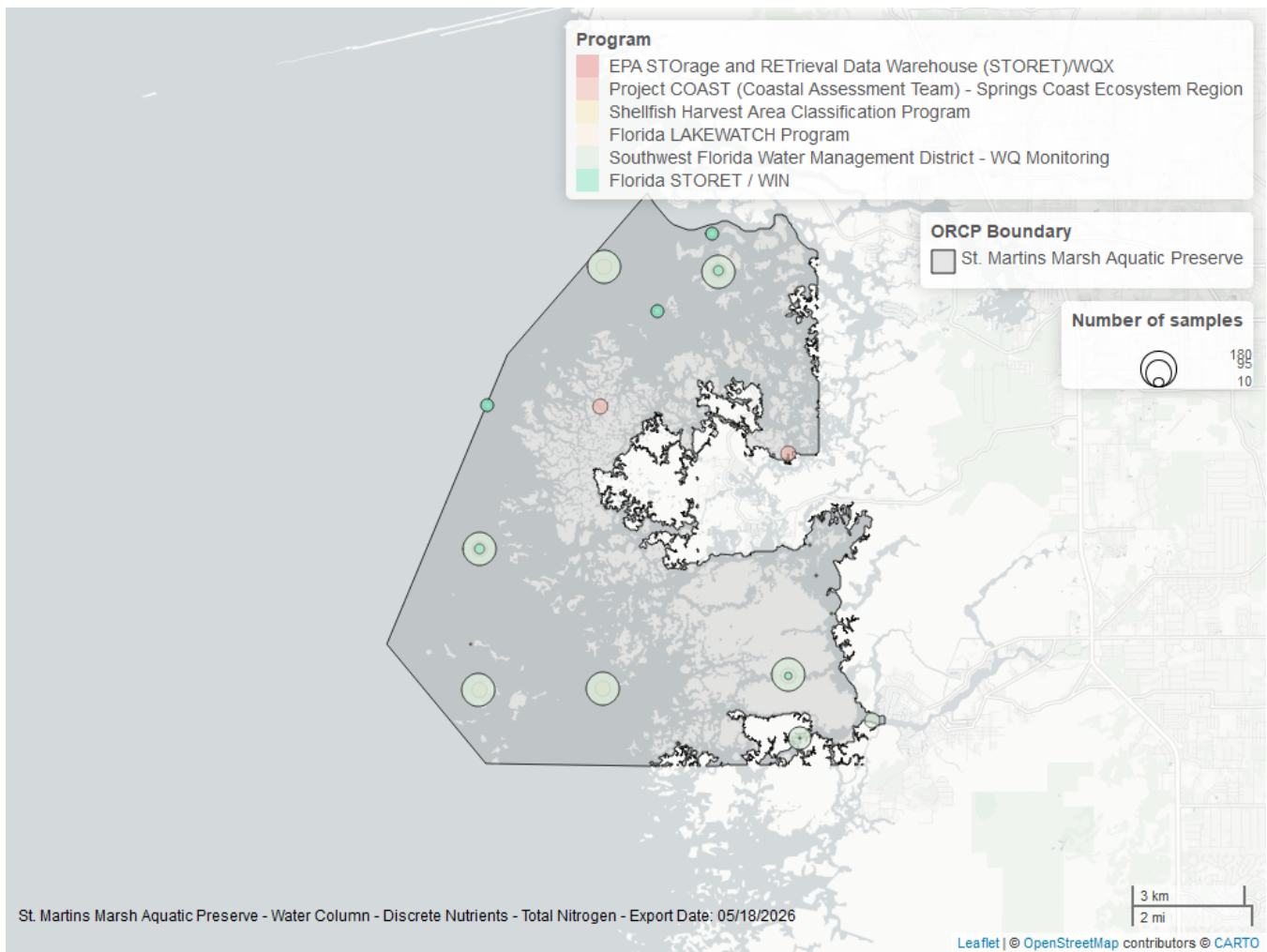


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
479	1131	1996	2025
5008	268	2022	2025
5002	119	2000	2021
540	21	2017	2019
514	6	2000	2000
103	1	2015	2015

Program names:

- 479 - Southwest Florida Water Management District - Water Quality Monitoring²
5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region¹
5002 - Florida STORET / WIN³
540 - Shellfish Harvest Area Classification Program⁴
514 - Florida LAKEWATCH Program⁵
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

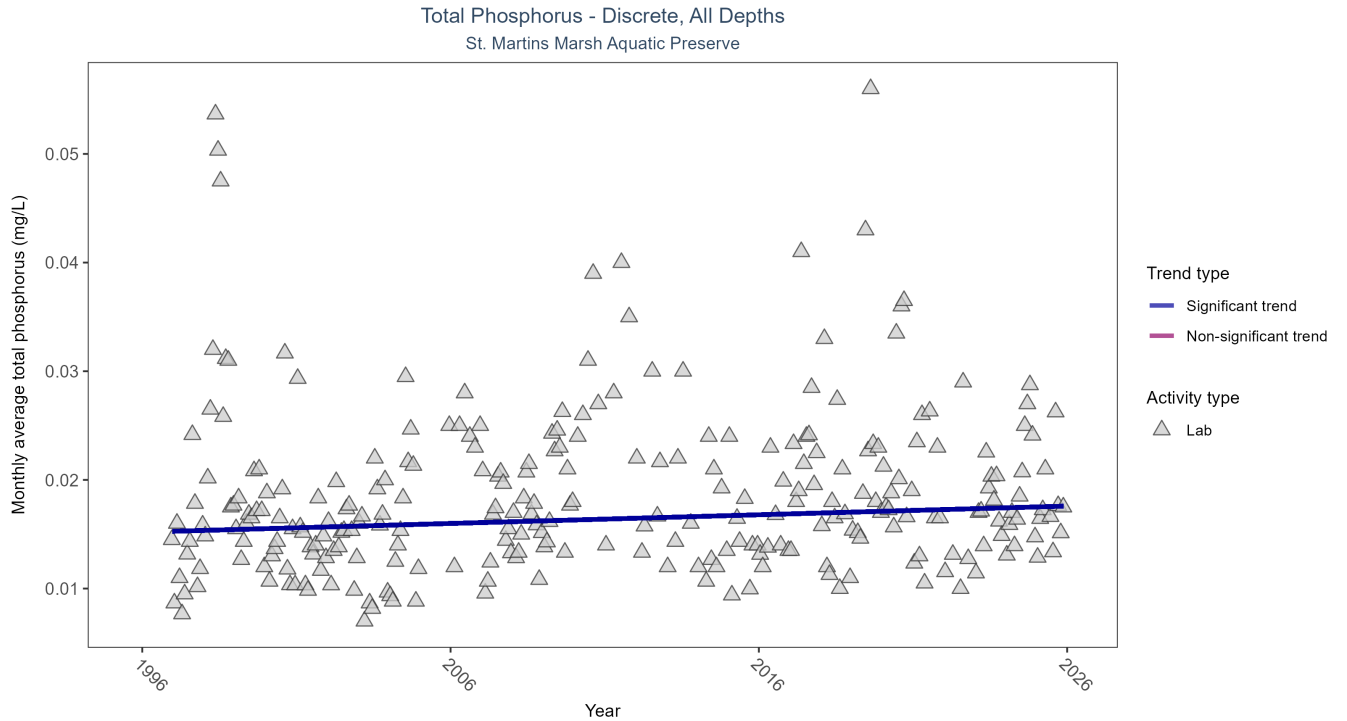


Figure 15: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1526	30	1996 - 2025	0.015	0.1046	0.0152	0.0001	0.0375

Monthly average total phosphorus increased by less than 0.01 mg/L per year.

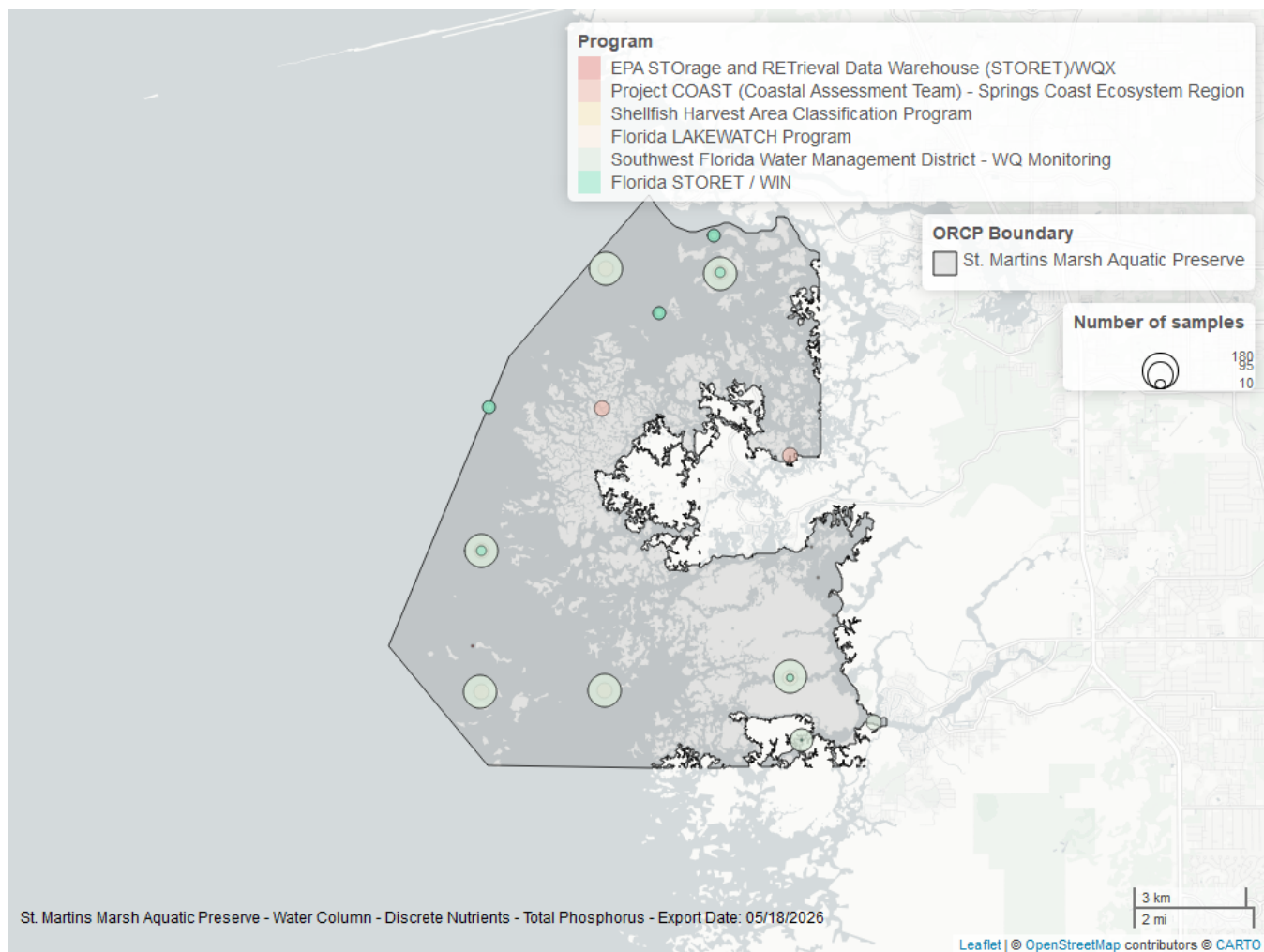


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
479	1129	1996	2025
5008	260	2022	2025
5002	113	2012	2021
540	21	2017	2019
514	6	2000	2000
103	1	2015	2015

Program names:

- 479 - Southwest Florida Water Management District - Water Quality Monitoring²
5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region¹
5002 - Florida STORET / WIN³
540 - Shellfish Harvest Area Classification Program⁴
514 - Florida LAKEWATCH Program⁵
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

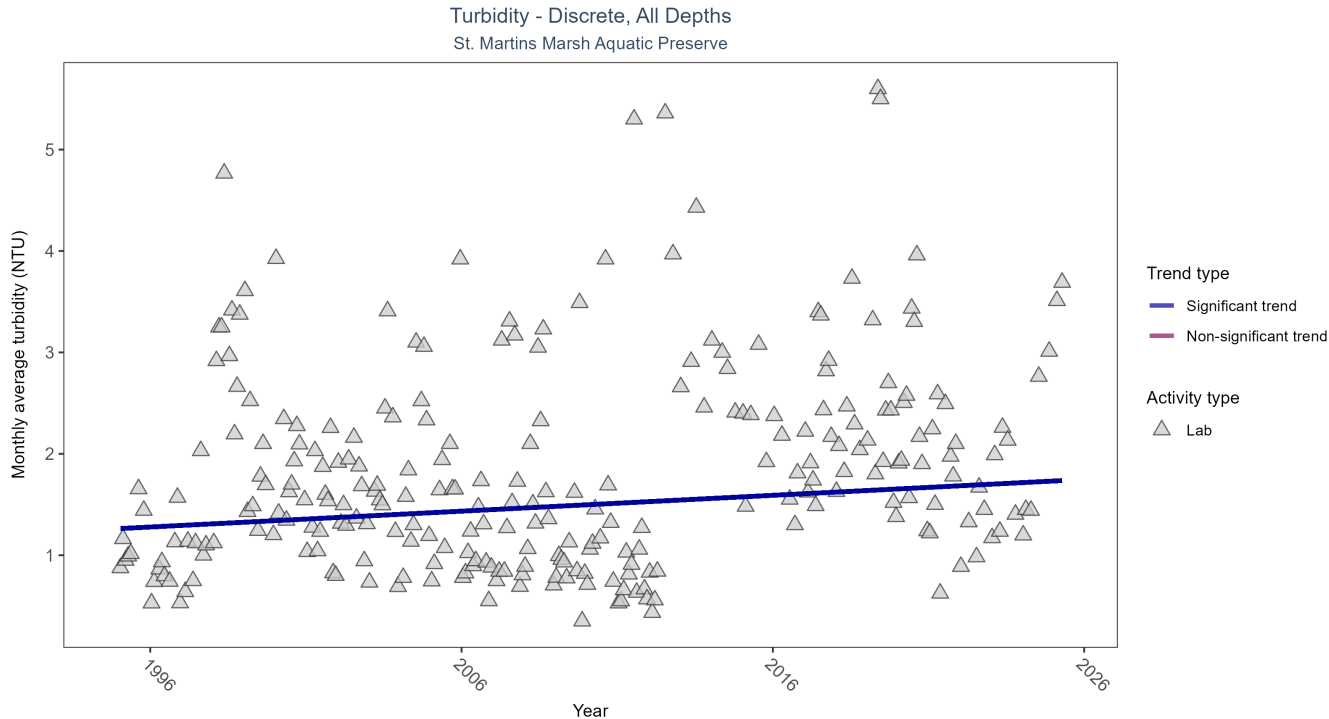


Figure 17: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	3157	31	1995 - 2025	1.1	0.1158	1.2625	0.0156	0.0101

Monthly average turbidity increased by 0.02 NTU per year, indicating a decrease in water clarity.

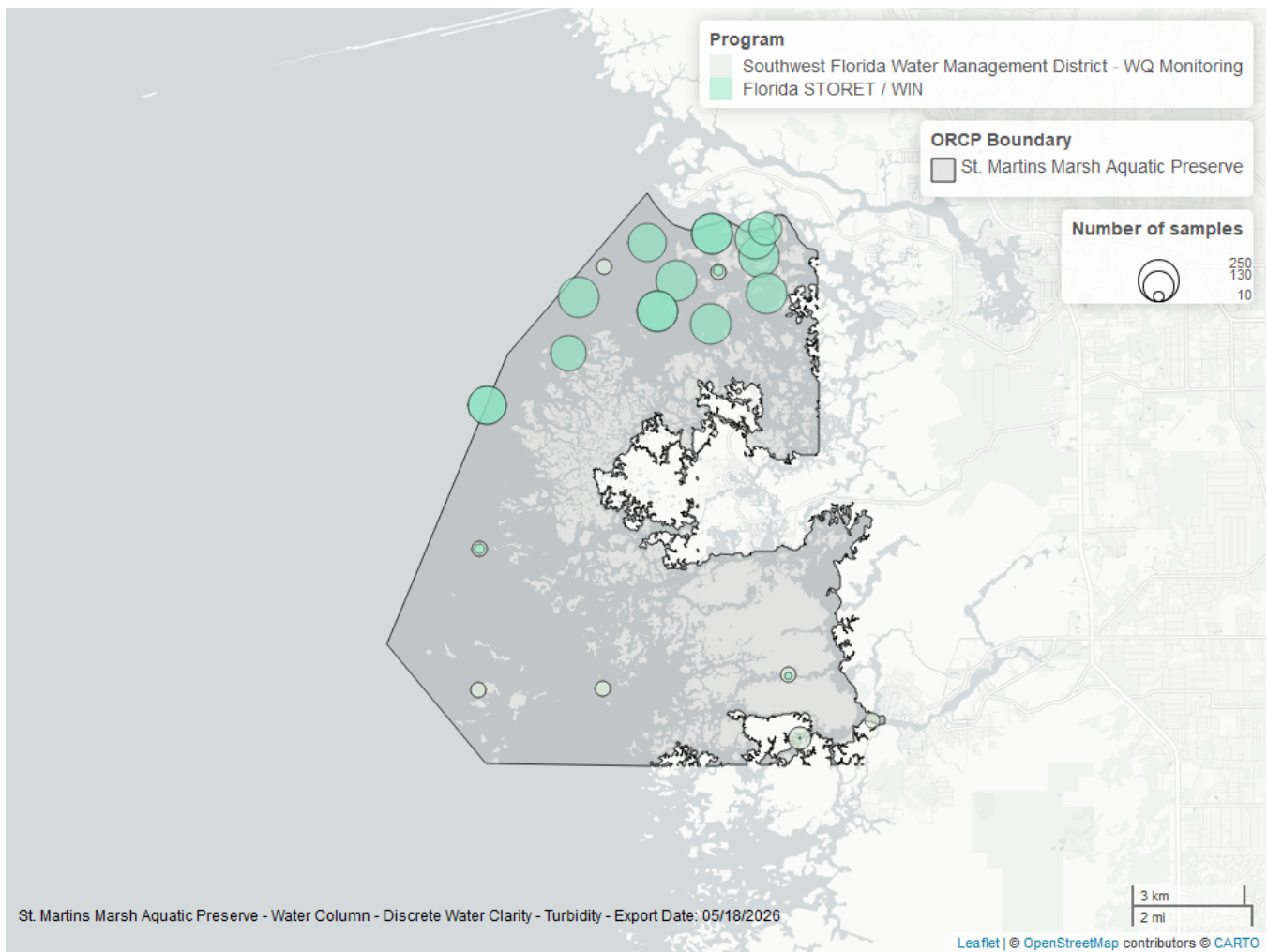


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	2819	1995	2021
479	340	2005	2025

Program names:

5002 - Florida STORET / WIN³

479 - Southwest Florida Water Management District - Water Quality Monitoring²

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

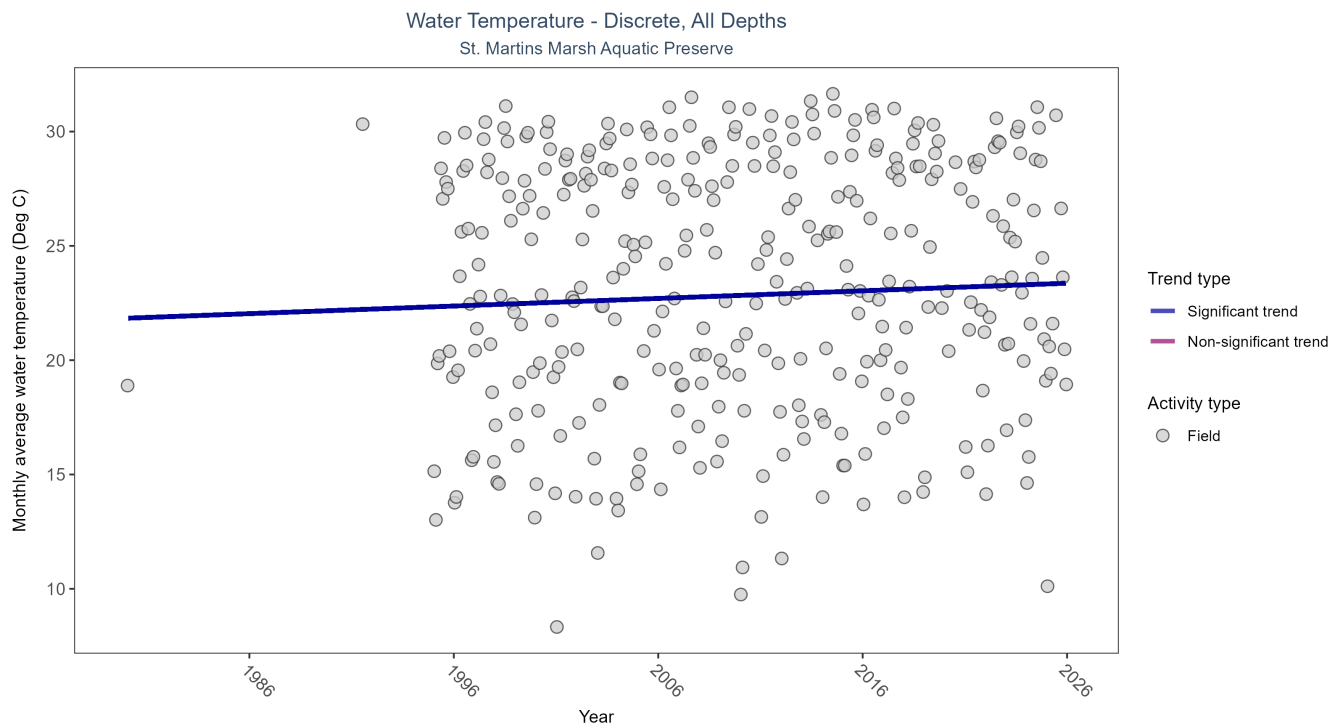


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	9155	33	1980 - 2025	22.7	0.1244	21.8356	0.0333	0.0013

Monthly average water temperature increased by 0.03°C per year.

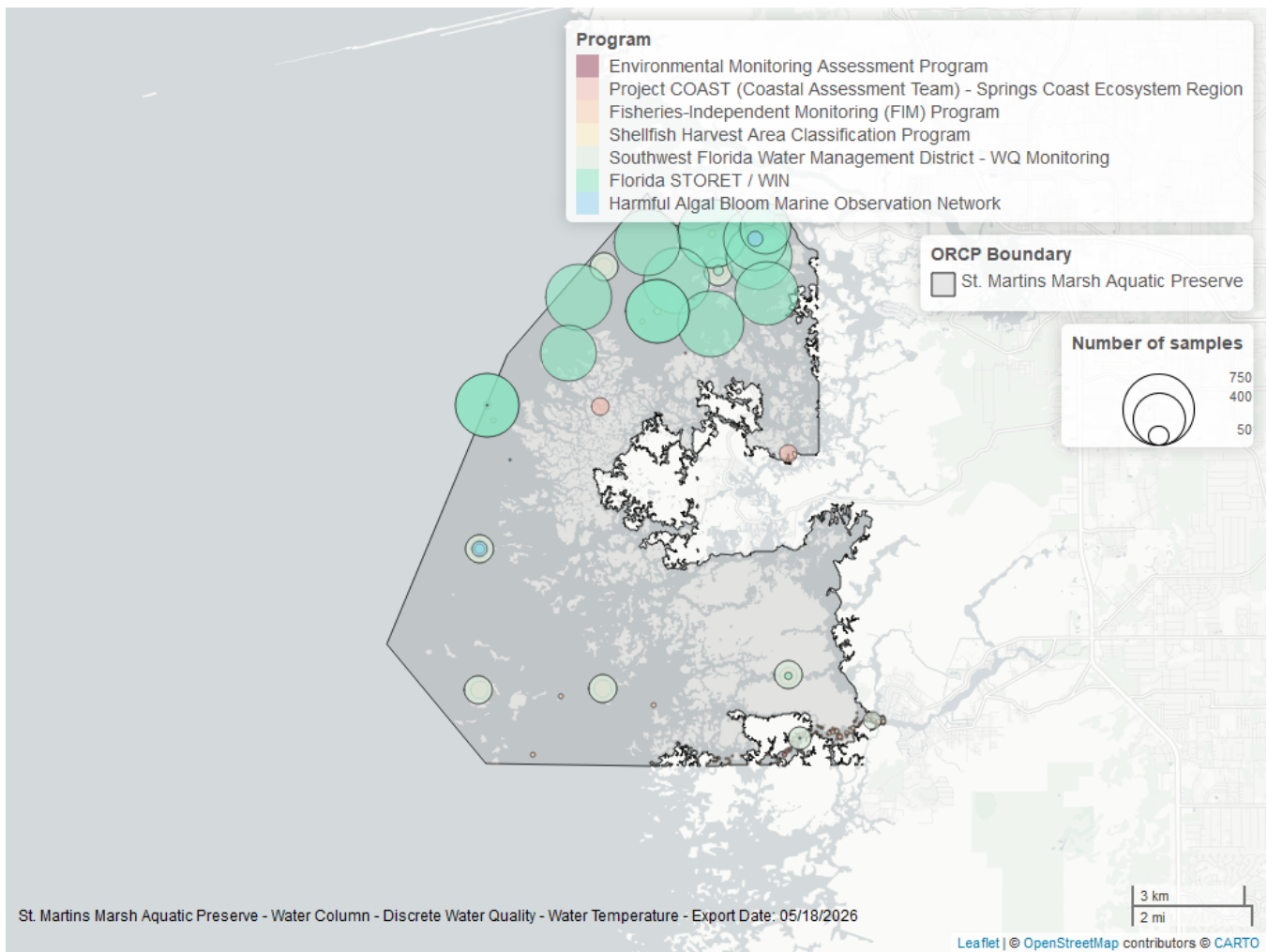


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	7535	1995	2021
479	879	1996	2025
5008	424	2021	2025
69	220	2003	2008
95	72	1980	2018
540	21	2017	2019
115	5	1991	2004

Program names:

5002 - Florida STORET / WIN³

479 - Southwest Florida Water Management District - Water Quality Monitoring²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region¹

69 - Fisheries-Independent Monitoring (FIM) Program⁹

95 - Harmful Algal Bloom Marine Observation Network¹⁰

540 - Shellfish Harvest Area Classification Program⁴
115 - Environmental Monitoring Assessment Program⁸

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in St. Martins Marsh Aquatic Preserve

Table 26: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
471	BBSCH	8	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity

Program names:

[471](#) - Big Bend Seagrasses Aquatic Preserves Continuous Water Quality Monitoring¹²

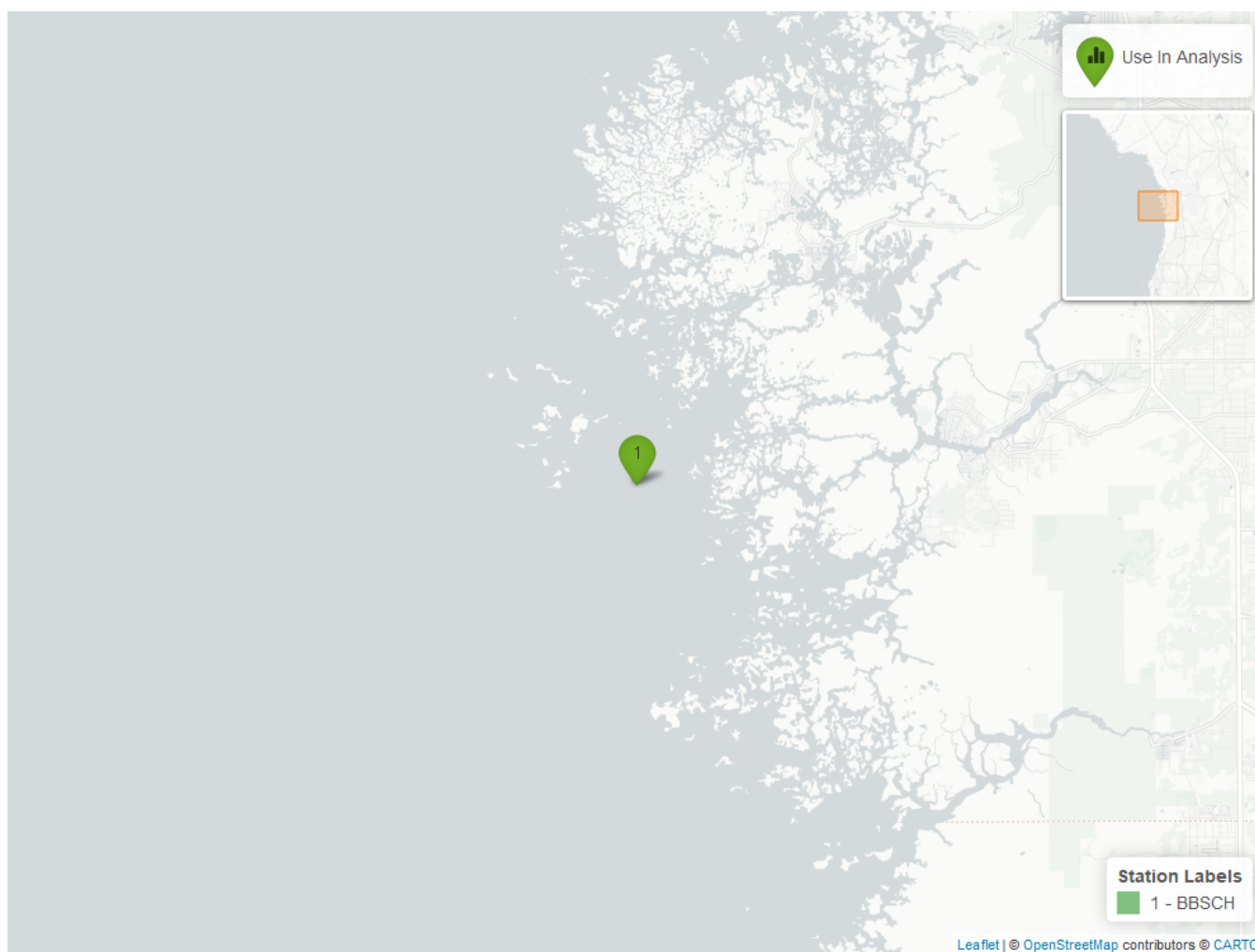


Figure 21: Map showing continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

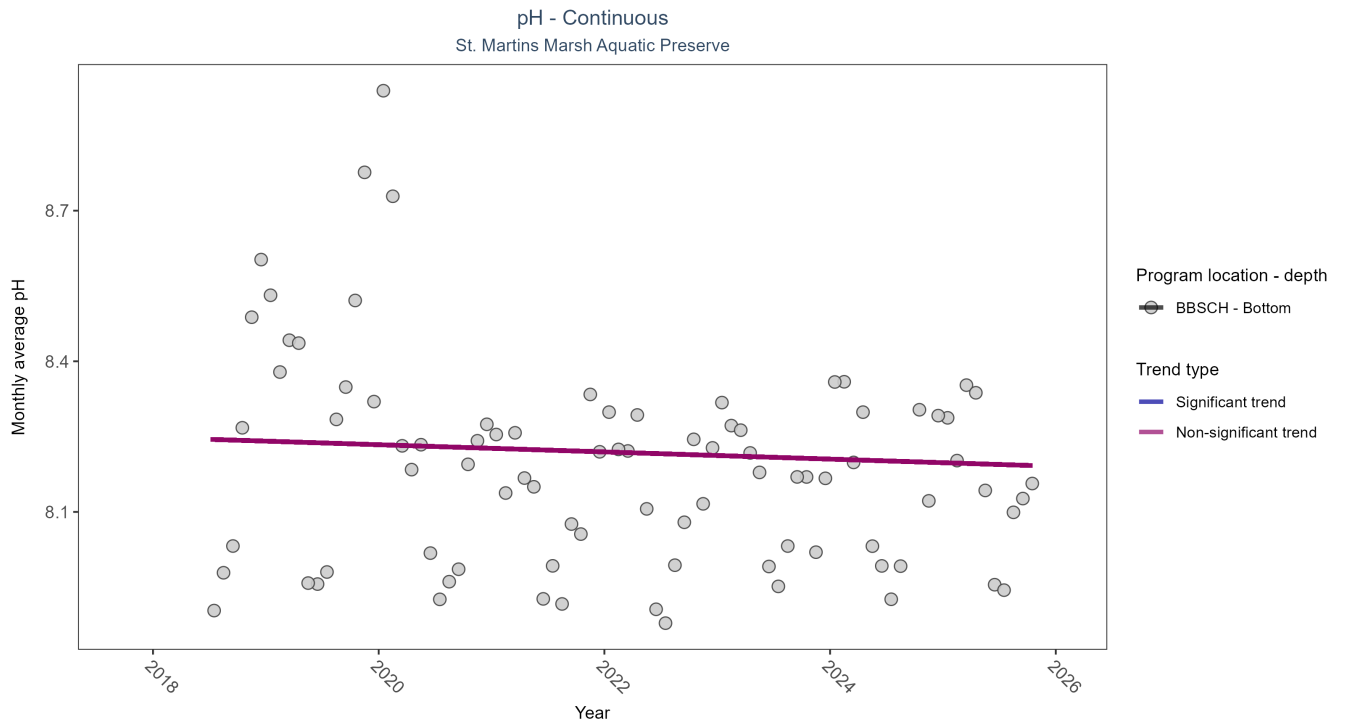


Figure 22: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 27: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	No significant trend	229419	8	2018 - 2025	8.2	-0.1	8.25	-0.01	0.29

No detectable change in monthly average pH was observed at one location.

Dissolved Oxygen Saturation - Continuous

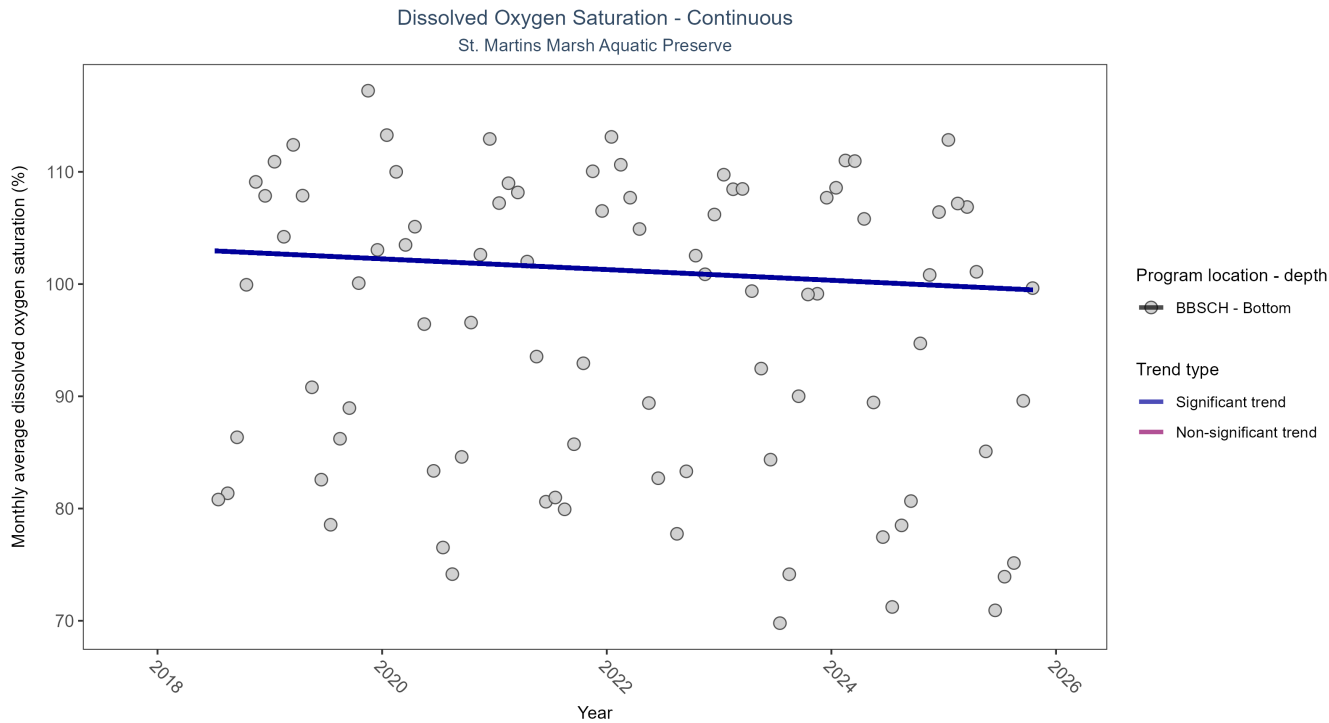


Figure 23: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 28: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly decreasing trend	227936	8	2018 - 2025	97.7	-0.26	103.2	-0.48	0

At one program location, monthly average dissolved oxygen saturation decreased by 0.48% per year.

Dissolved Oxygen - Continuous

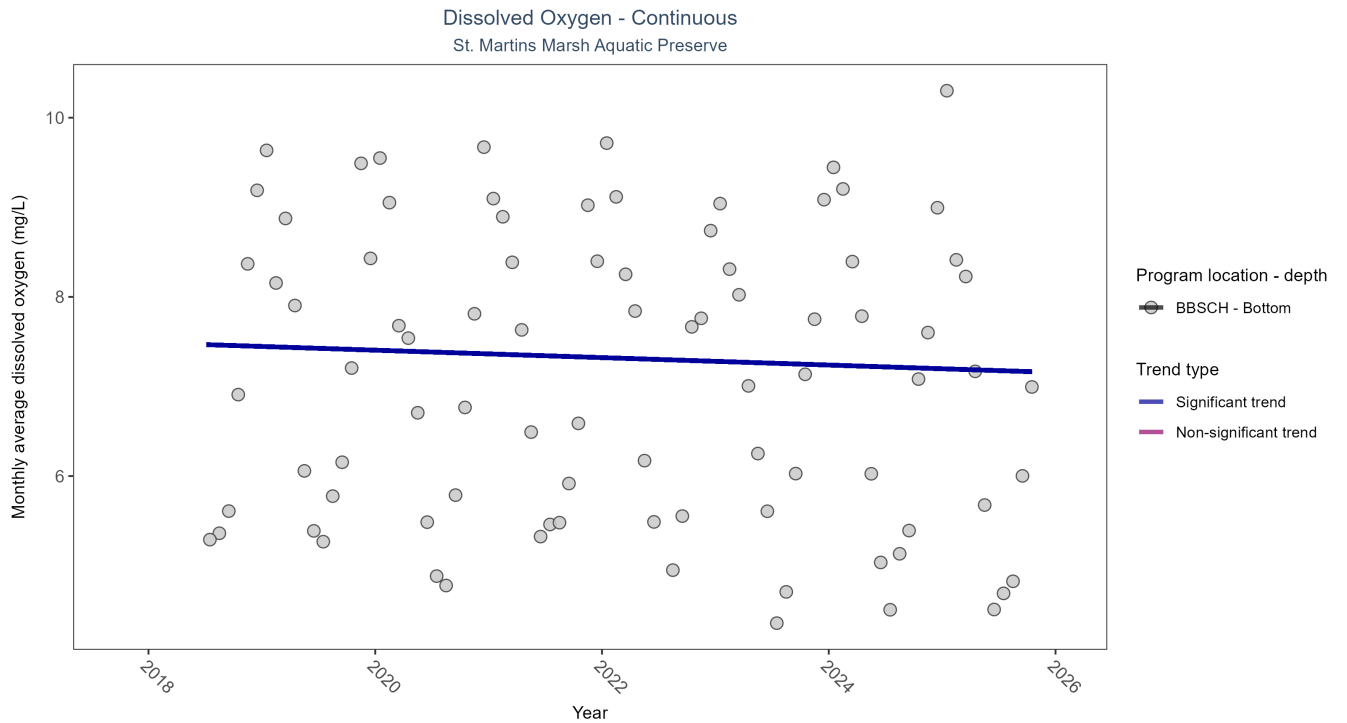


Figure 24: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly decreasing trend	227936	8	2018 - 2025	7.2	-0.21	7.49	-0.04	0.02

At one program location, monthly average dissolved oxygen decreased by 0.04 mg/L per year.

Turbidity - Continuous

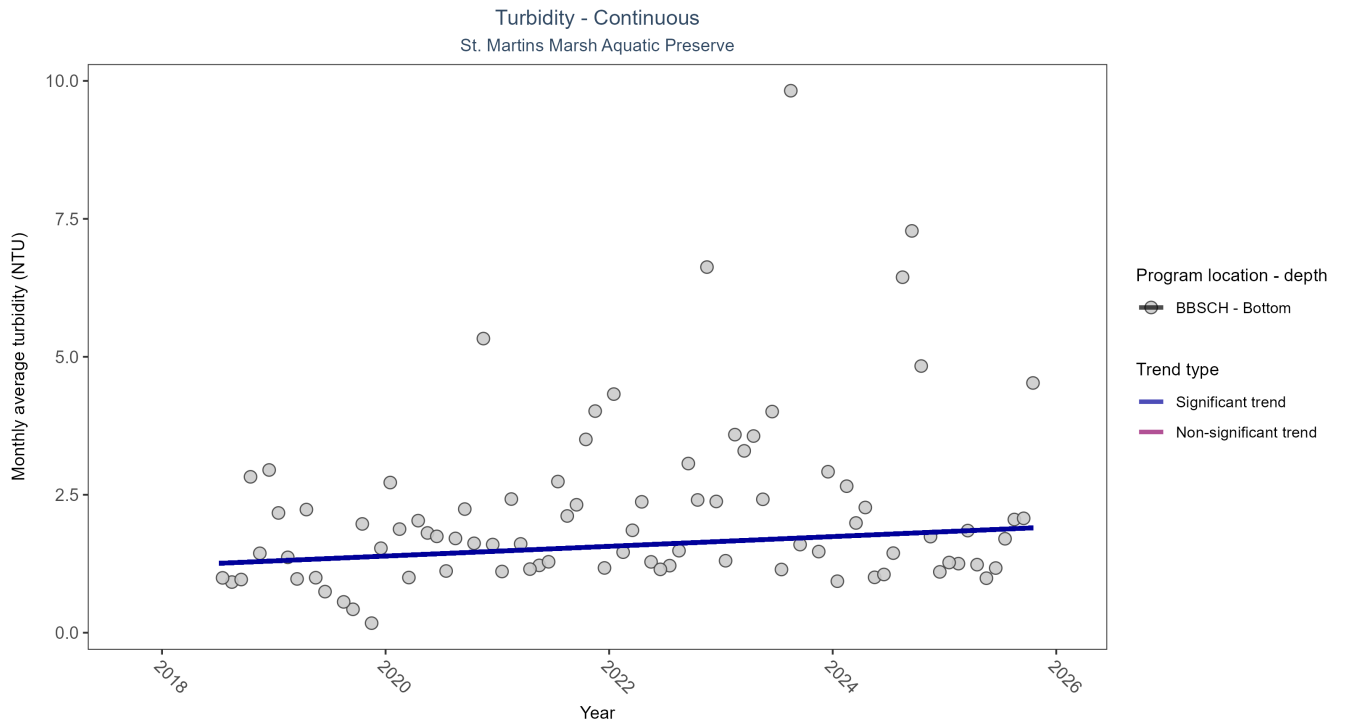


Figure 25: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly increasing trend	204374	8	2018 - 2025	1	0.19	1.21	0.09	0.03

At one program location, monthly average turbidity increased by 0.09 NTU per year.

Water Temperature - Continuous

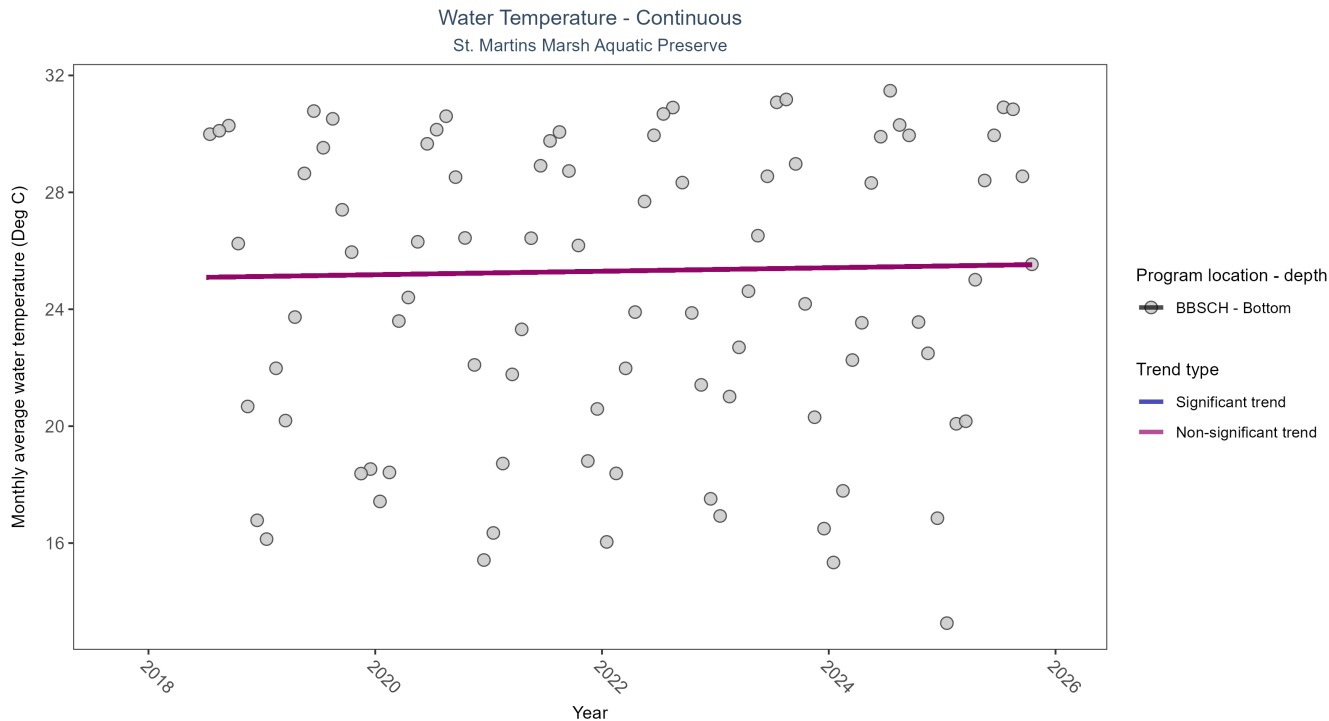


Figure 26: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Water Temperature - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	No significant trend	237837	8	2018 - 2025	25.3	0.06	25.06	0.06	0.49

No detectable change in monthly average water temperature was observed at one location.

Salinity - Continuous

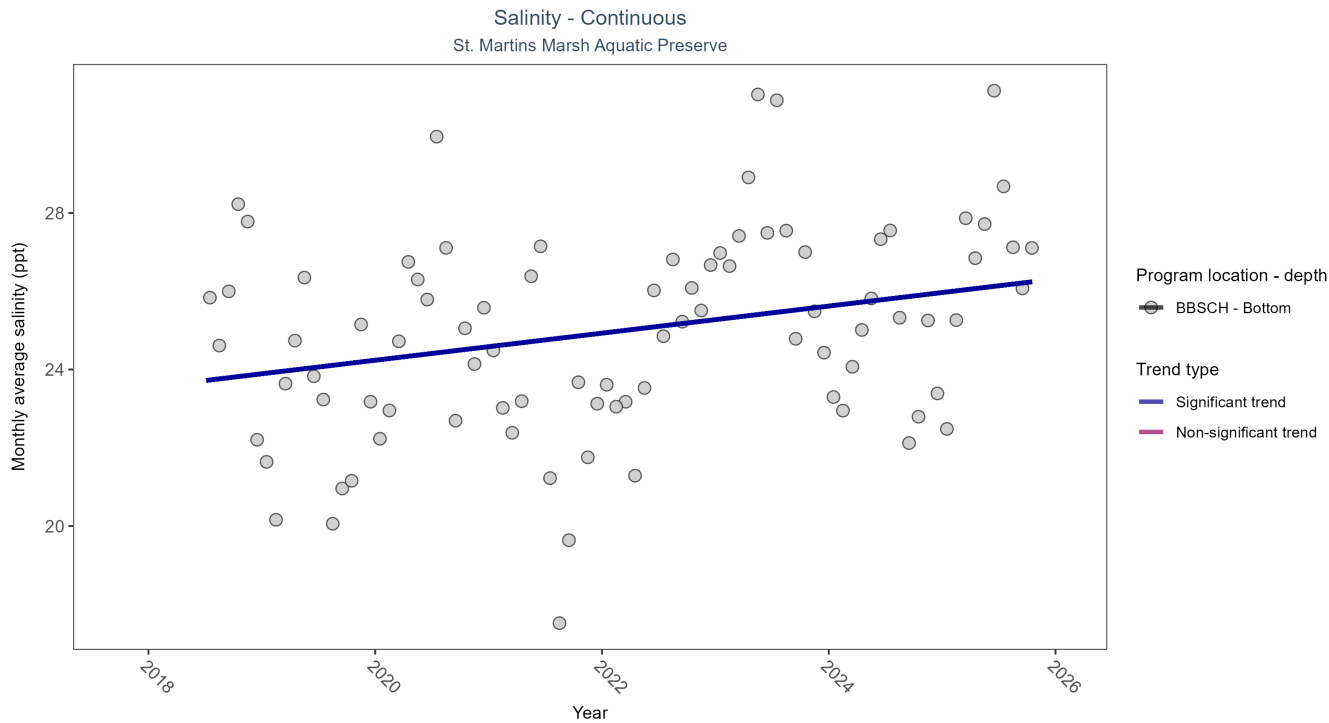


Figure 27: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Salinity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly increasing trend	235777	8	2018 - 2025	25.2	0.29	23.55	0.35	0

At one program location, monthly average salinity increased by 0.35 ppt per year.

Specific Conductivity - Continuous

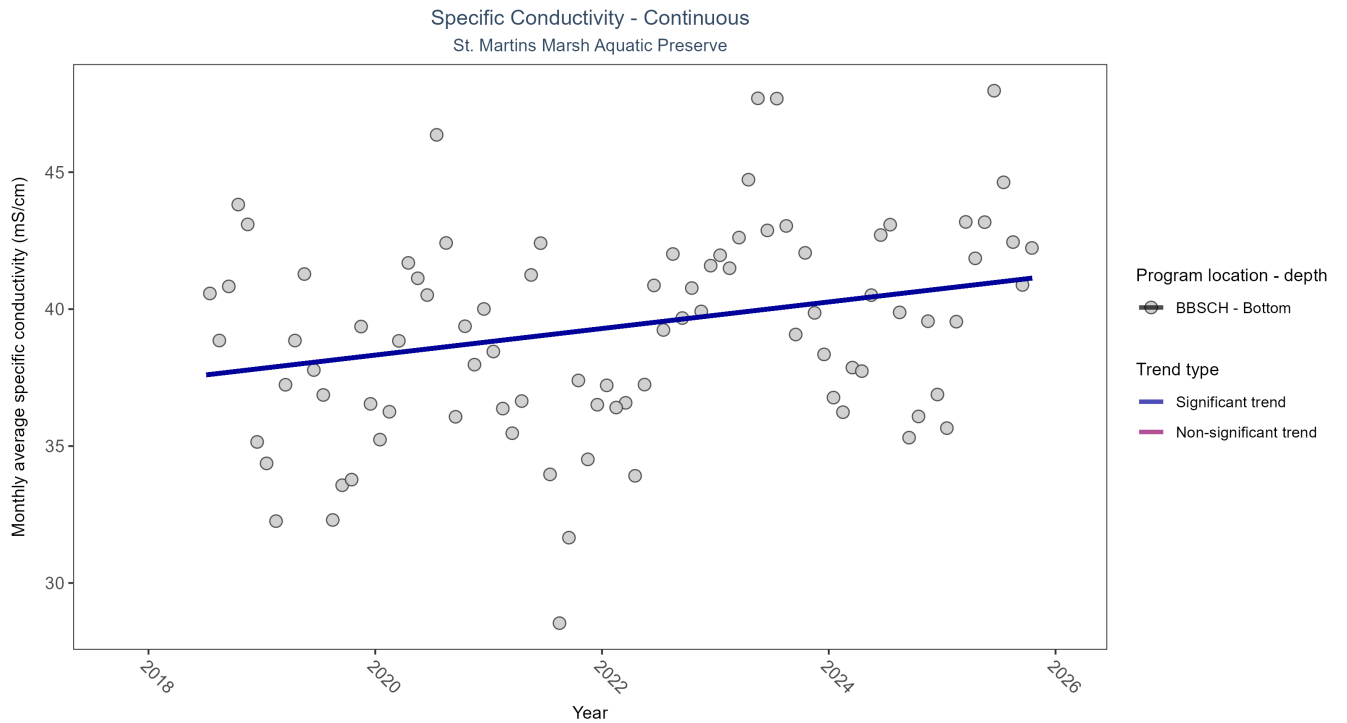


Figure 28: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly increasing trend	234051	8	2018 - 2025	39.55	0.27	37.35	0.49	0

At one program location, monthly average specific conductivity increased by 0.49 mS/cm per year.

Submerged Aquatic Vegetation

The data file used is: **All_SAV_Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (SAV) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of SAV inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover.”

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida's estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila* spp. are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson's seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson's seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

St. Martins Marsh Aquatic Preserve
SAV Percent Cover - Sample Locations

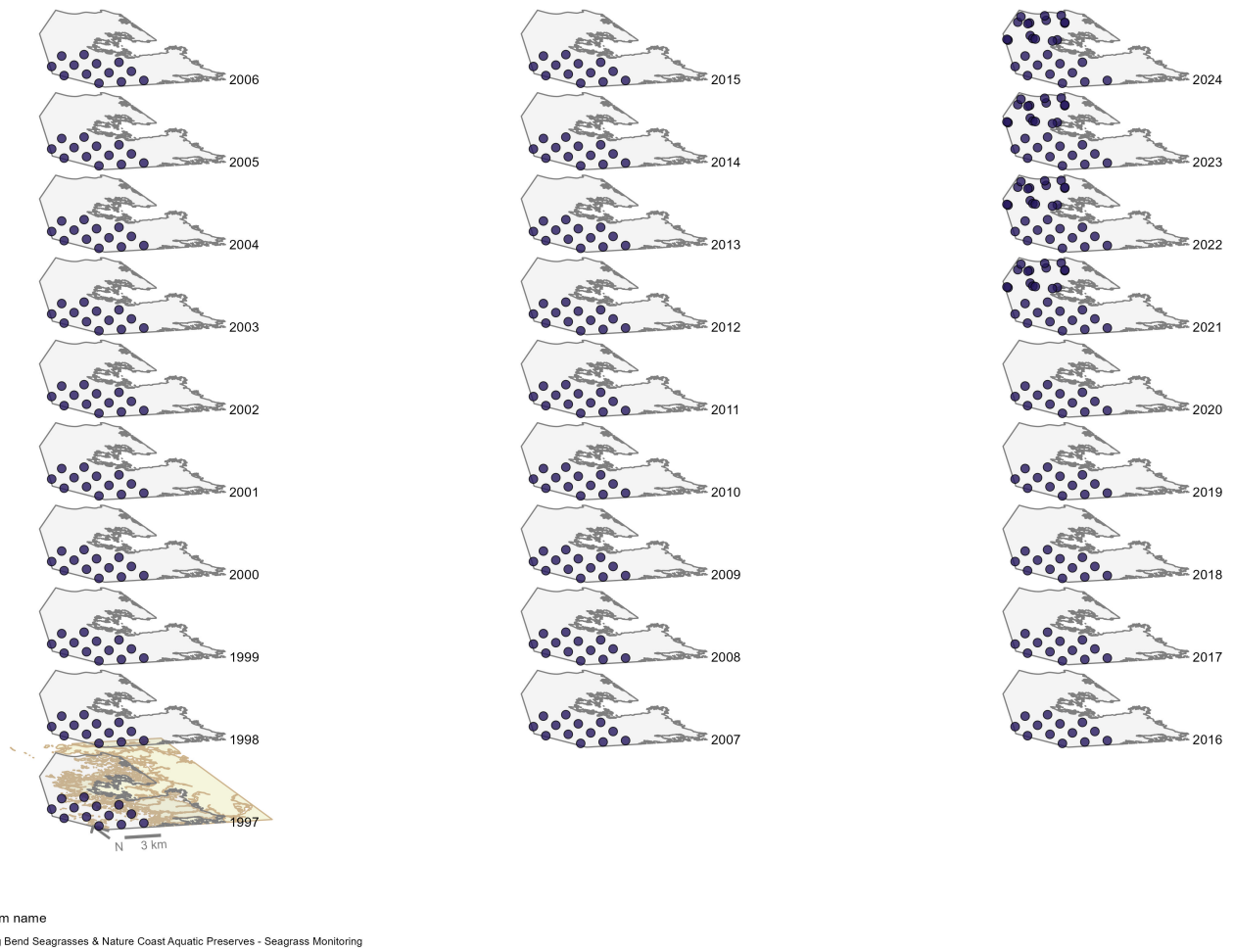


Figure 29: Maps showing the temporal scope of SAV sampling sites within the boundaries of *St. Martins Marsh Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Sampling locations by Program:

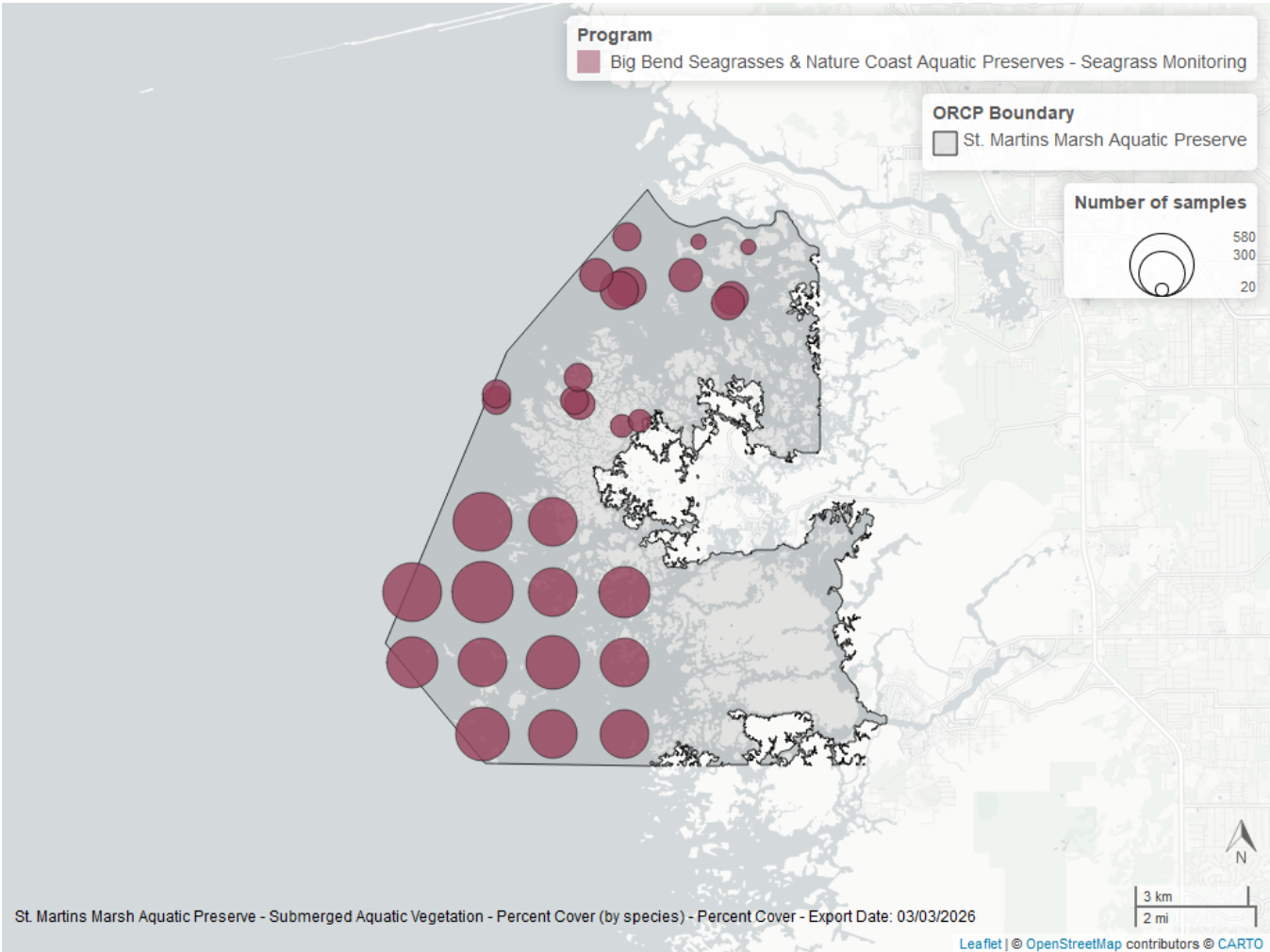


Figure 30: Map showing SAV sampling sites within the boundaries of *St. Martins Marsh Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 34: Program Information for Submerged Aquatic Vegetation

<i>ProgramID</i>	<i>N-Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>method</i>	<i>Sample Locations</i>
560	5968	1997	2025	Modified Braun Blanquet	29
560	1676	2021	2025	Percent Cover	29

Program names:

[560](#) - Big Bend Seagrasses & Nature Coast Aquatic Preserves - Seagrass Monitoring¹¹

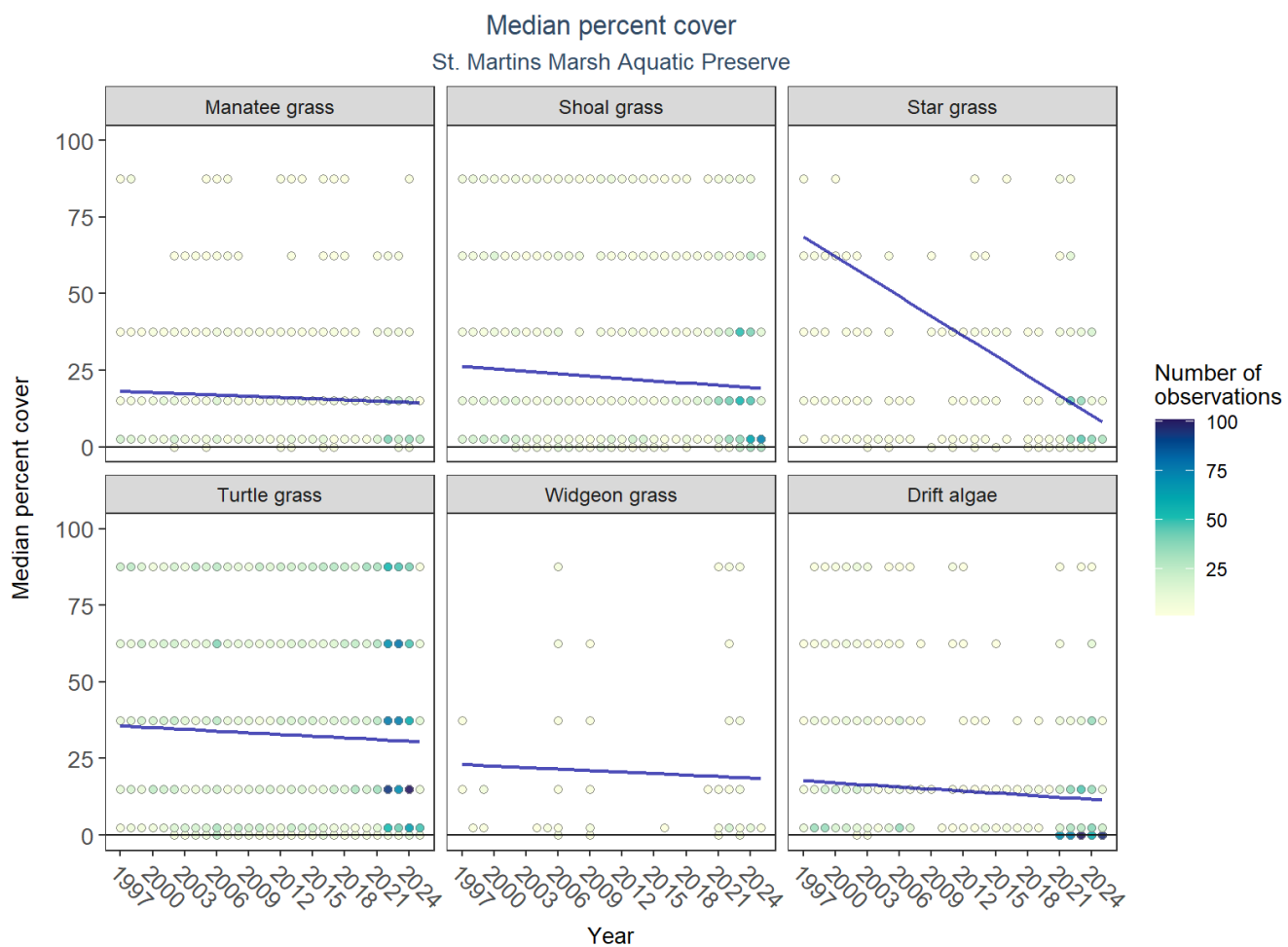


Figure 31: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

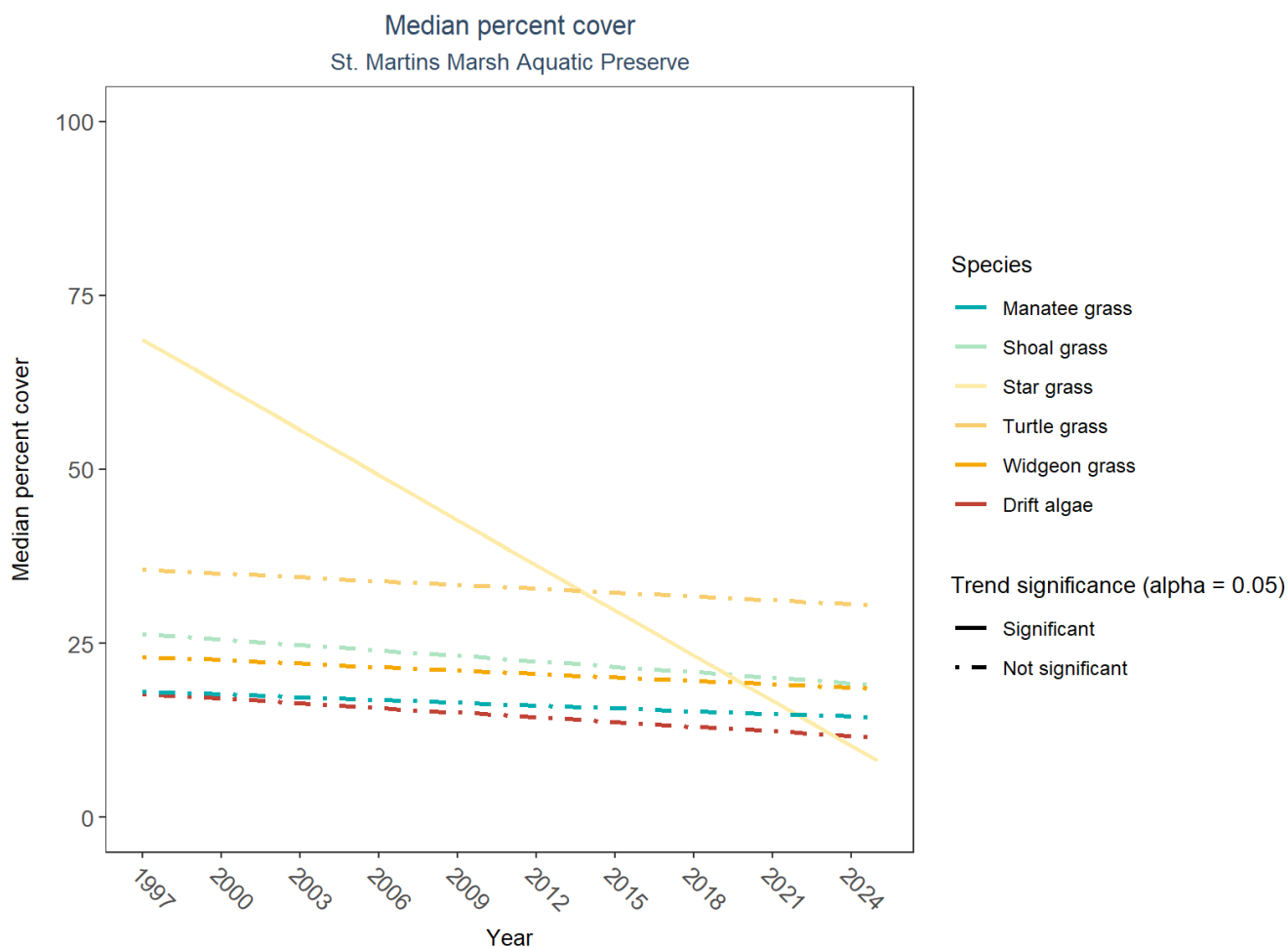


Figure 32: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 35: Percent Cover Trend Analysis for St. Martins Marsh Aquatic Preserve

<i>Species</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>LME Intercept</i>	<i>LME Slope</i>	<i>P</i>
Drift algae	No detectable trend	1997 - 2025	18.42305	-0.2240032	0.1034290
Shoal grass	No detectable trend	1997 - 2025	27.13598	-0.2614241	0.1437117
Star grass	Decreasing trend	1997 - 2025	75.14317	-2.1621449	0.0447030
Widgeon grass	No detectable trend	1997 - 2025	23.54146	-0.1630198	0.6640937
Manatee grass	No detectable trend	1997 - 2025	18.49436	-0.1341244	0.5112513
Turtle grass	No detectable trend	1997 - 2025	36.16267	-0.1835963	0.2490882

An annual decrease in percent cover was observed for star grass (-2.16%). Manatee grass, shoal grass, turtle grass, widgeon grass, and drift algae showed no detectable change in percent cover.

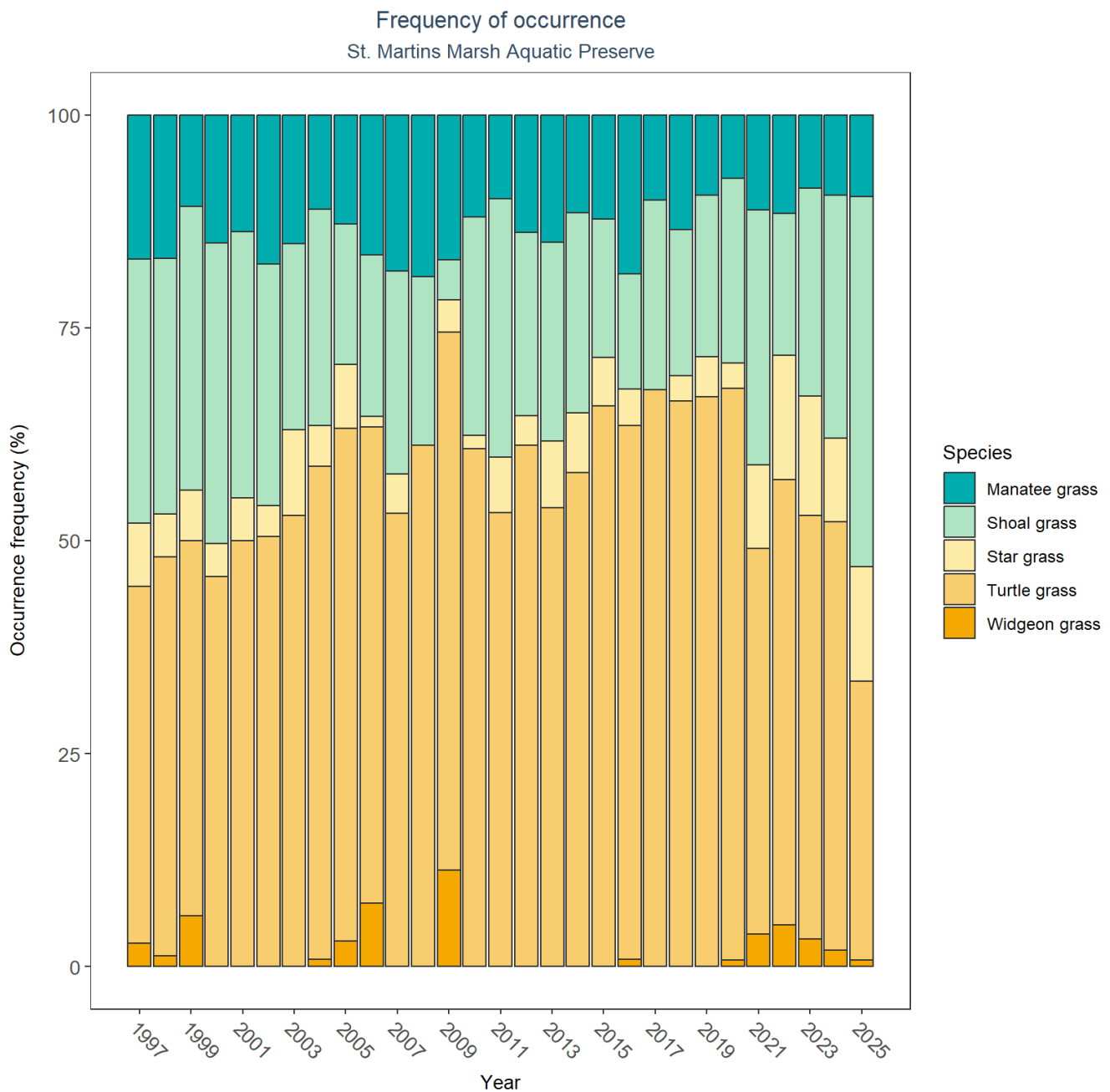


Figure 33: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for St. Martins Marsh Aquatic Preserve within the SAV_WC_Report:

- Colored Dissolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- pH
- Salinity
- Secchi Depth

- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: [DRAFT_SAV_WC_Report.pdf](#)

Nekton

The data file used is: **All_NEKTON_Parameters-2026-Jun-04.txt**

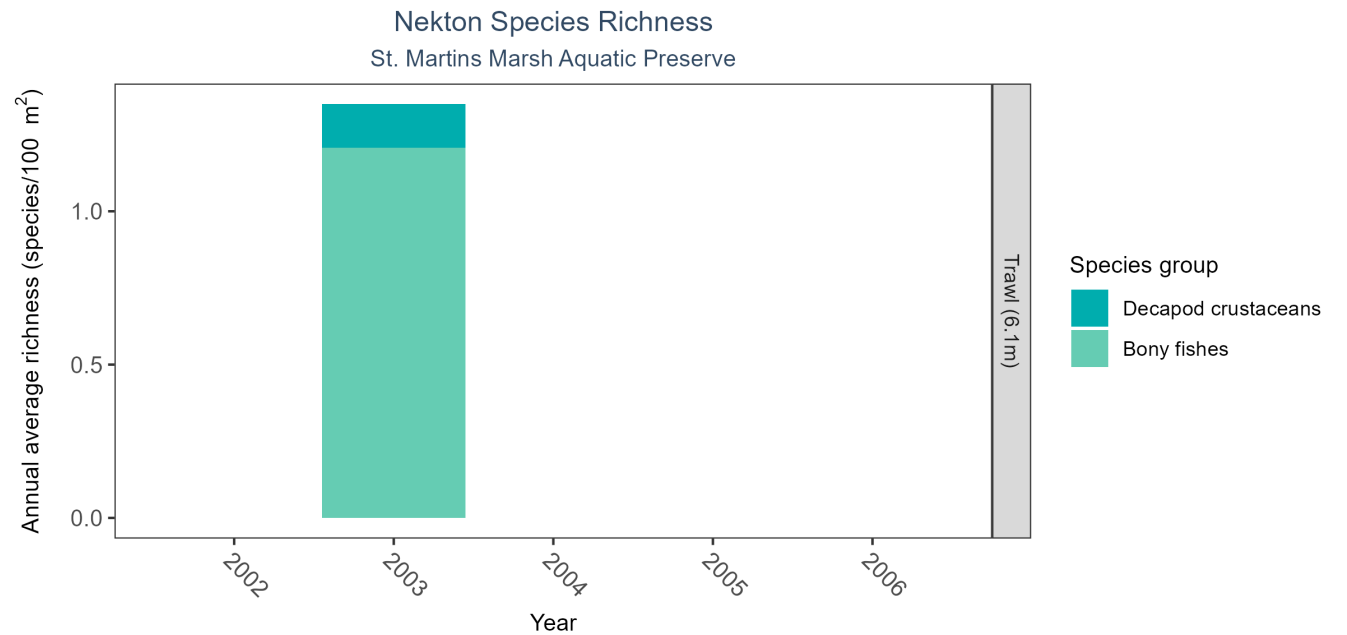


Figure 34: Bar graph(s) of annual average nekton richness over time for species groups occurring in at least 1% of samples. The bar colors represent species groups including bony fishes, cartilaginous fishes, decapod crustaceans (e.g., shrimps, crabs, and lobsters), and cephalopods (e.g., squid). Gear types and sizes are indicated in the panel label.

Table 36: Nekton Species Richness

<i>Gear Type</i>	<i>No. of Samples</i>	<i>No. of Years with Data</i>	<i>Period of Record</i>	<i>Median No. of Taxa</i>	<i>Mean No. of Taxa</i>
Trawl (6.1 m)	2	1	2003 - 2003	0.67	0.67

The median annual number of taxa was 0.67 based on 2 observations collected by 6.1-meter trawl between 2003 and 2003.

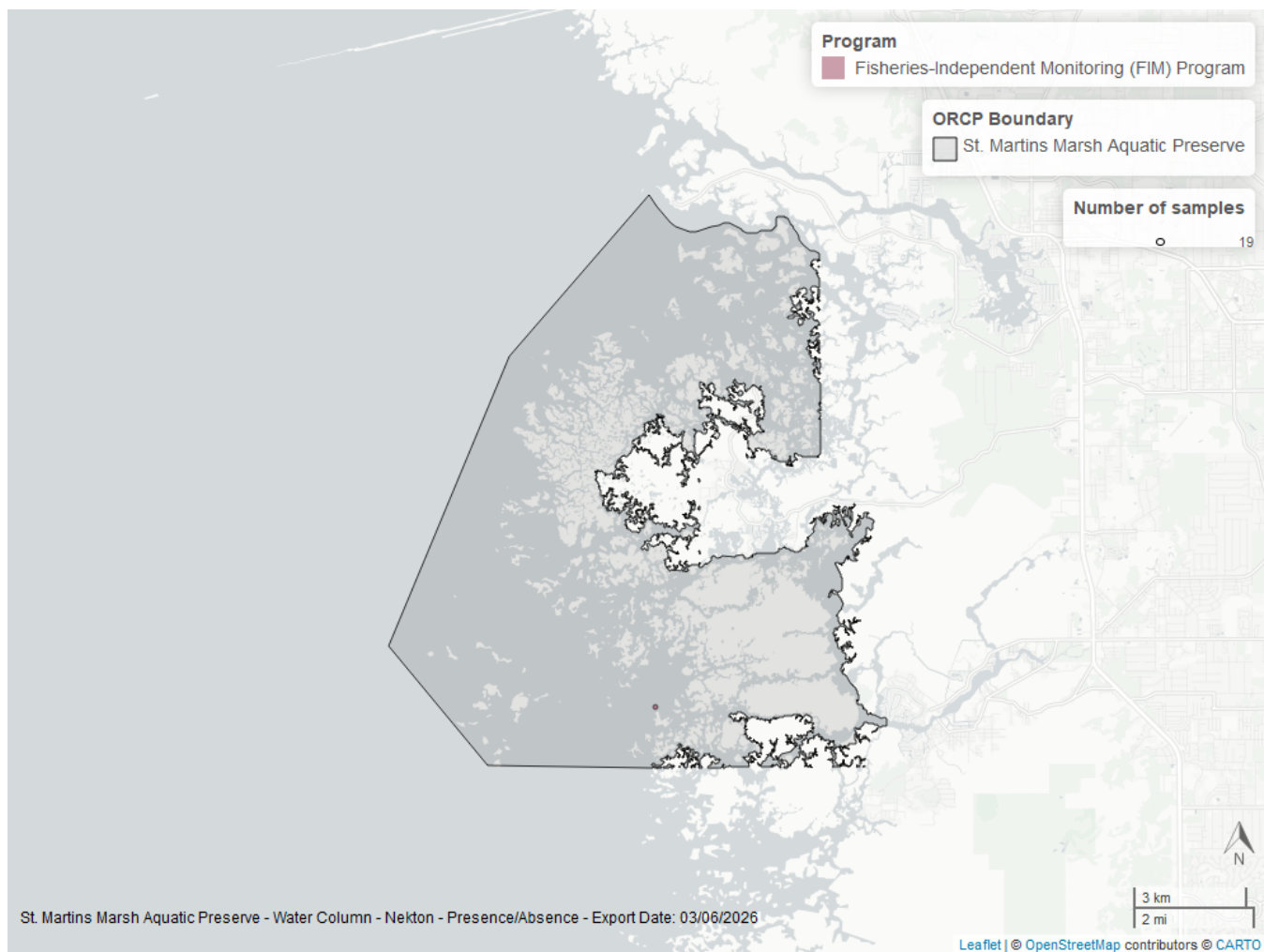


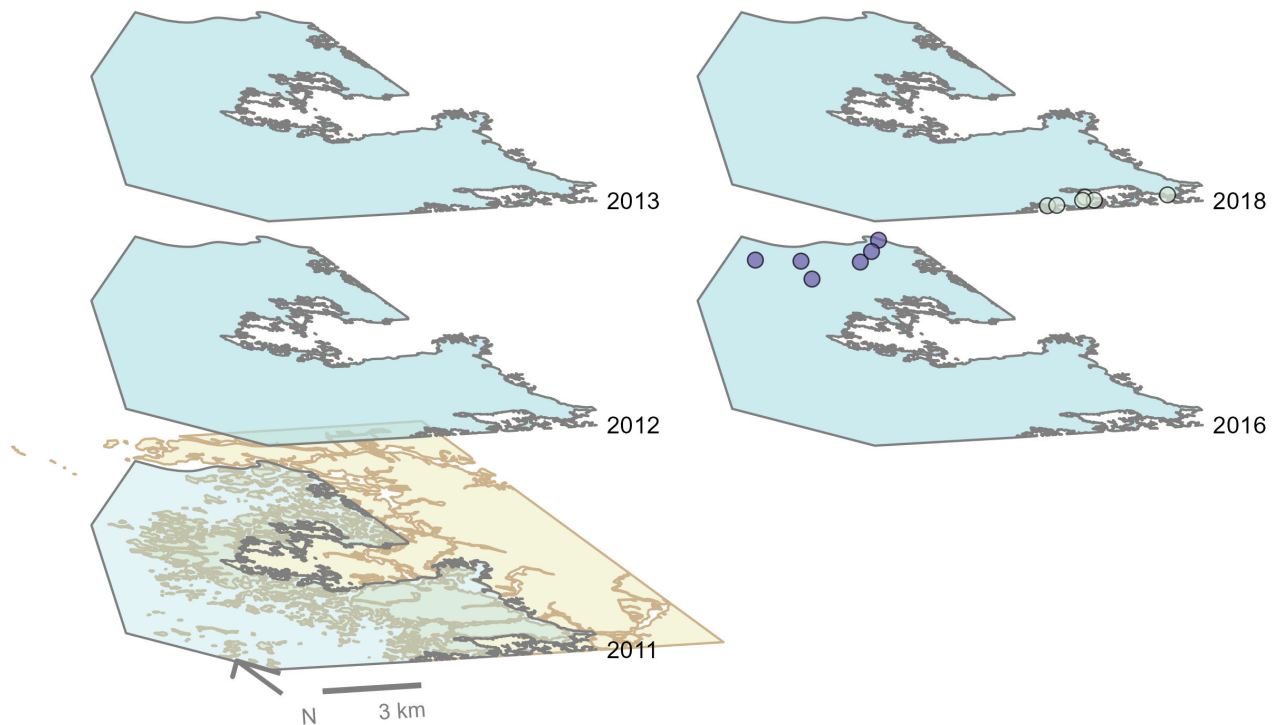
Figure 35: Map showing location of nekton sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Oyster

The data file used is: **All_OYSTER_Parameters-2026-Jun-04.txt**

St. Martins Marsh Aquatic Preserve

Oyster sample locations available in SEACAR by Program and Year



Program name

- Prehistoric baseline reveals substantial decline of oyster reef condition in a Gulf of Mexico conservation priority area - modern oyster data
- Assessment of Oysters and Barnacles Along the Chassahowitzka, Homosassa, and Lower Withlacoochee River Systems

Figure 36: Maps showing the temporal scope of oyster sampling sites within the boundaries of *St. Martins Marsh Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Density

For natural reefs, there was insufficient data to calculate a trend for density.

Natural

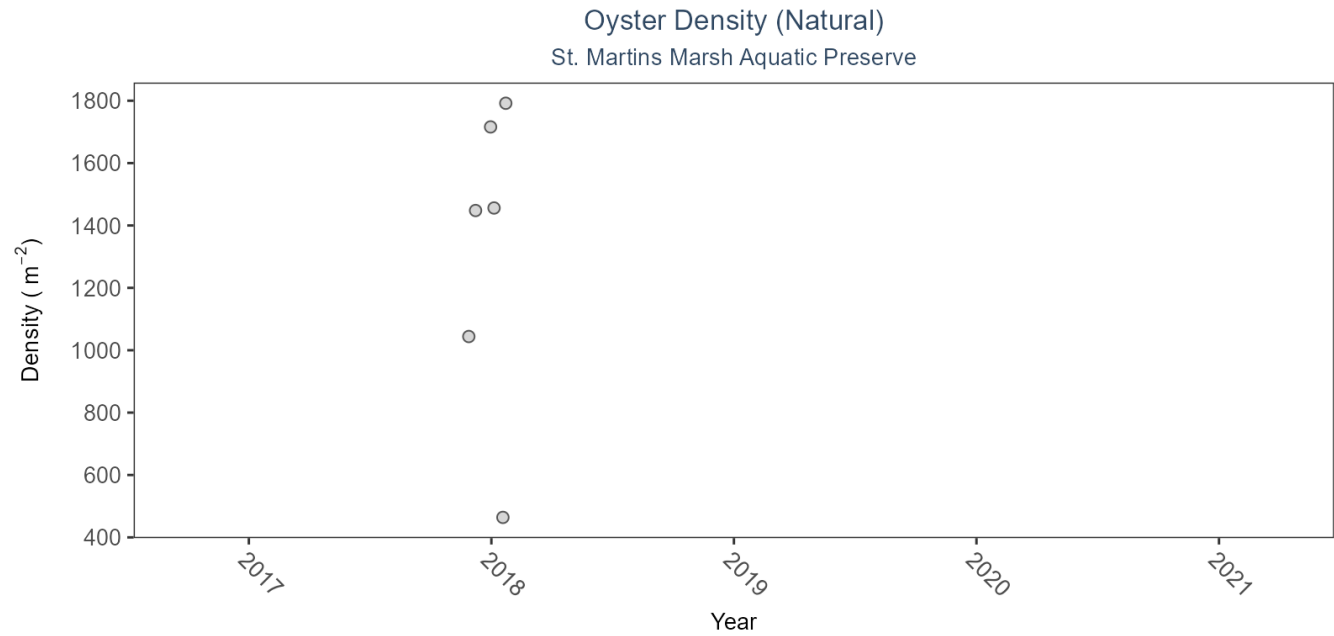


Figure 37: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 37: Model results for Oyster Density - Natural

Habitat Type	Shell Type	Statistical Trend	Period of Record	Estimate	Standard Error	Credible Interval
Natural	Live Oysters	Insufficient data	2018 - 2018	-	-	-

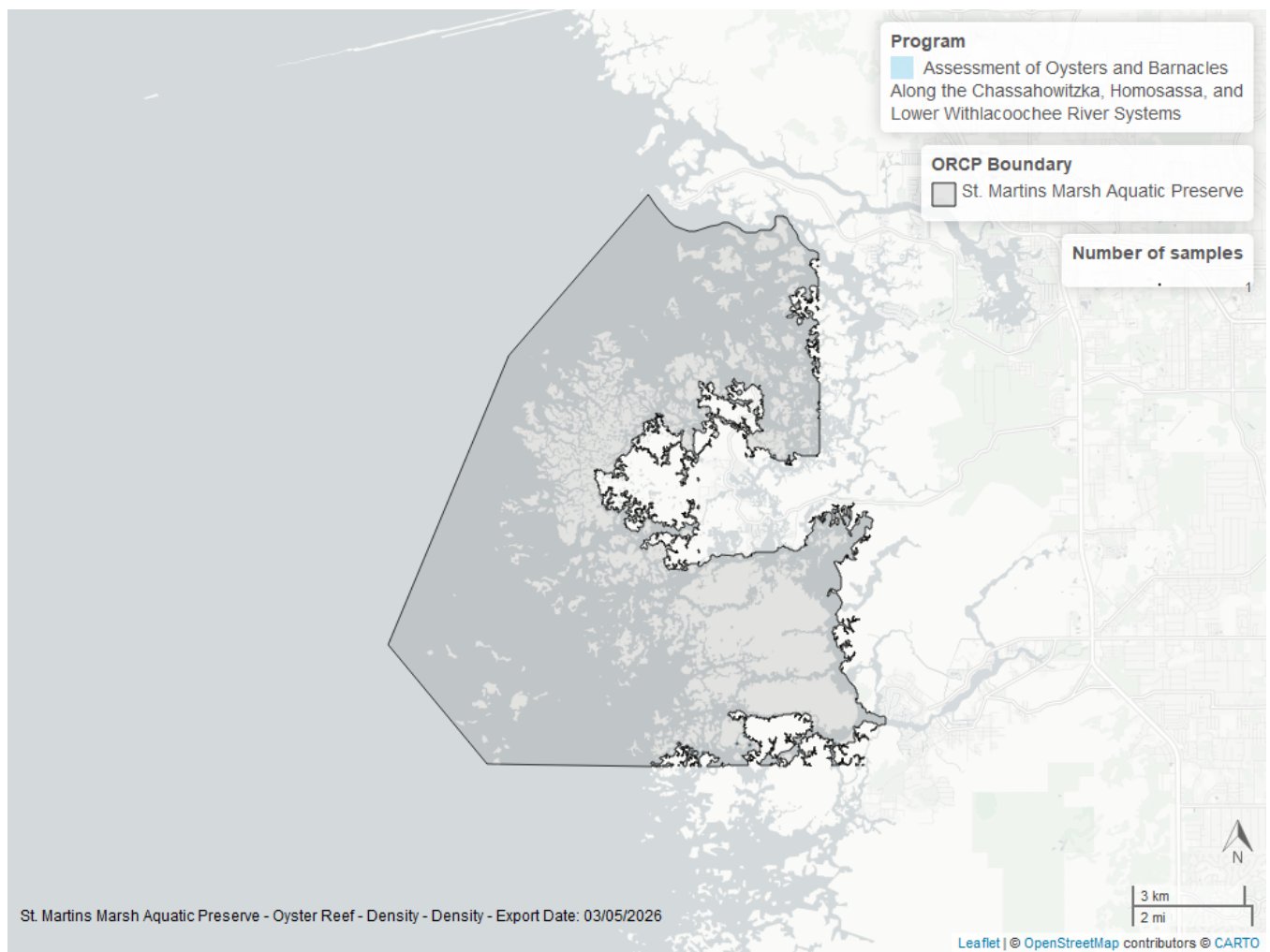


Figure 38: Map showing location of oyster density sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Percent Live

For natural reefs, there was insufficient data to calculate a trend for percent live cover.

Natural

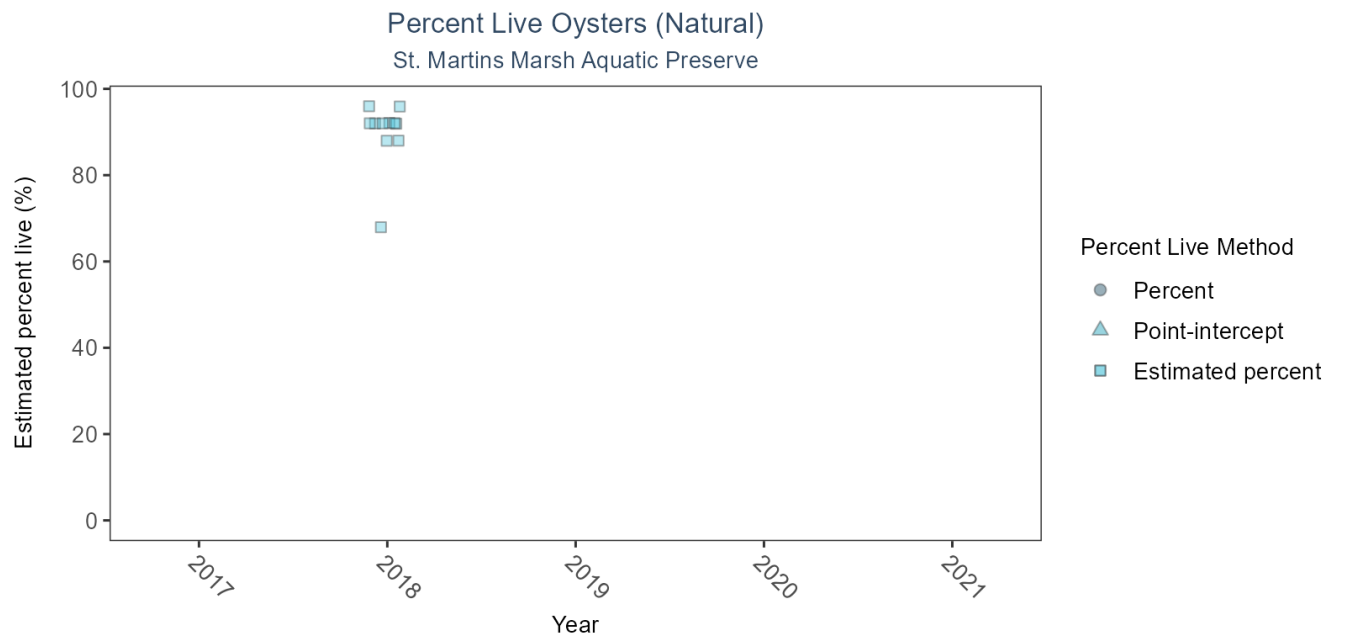


Figure 39: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 38: Model results for Oyster Percent Live - Natural

Habitat Type	Shell Type	Statistical Trend	Period of Record	Estimate	Standard Error	Credible Interval
Natural	Live Oysters	Insufficient data	2018 - 2018	-	-	-

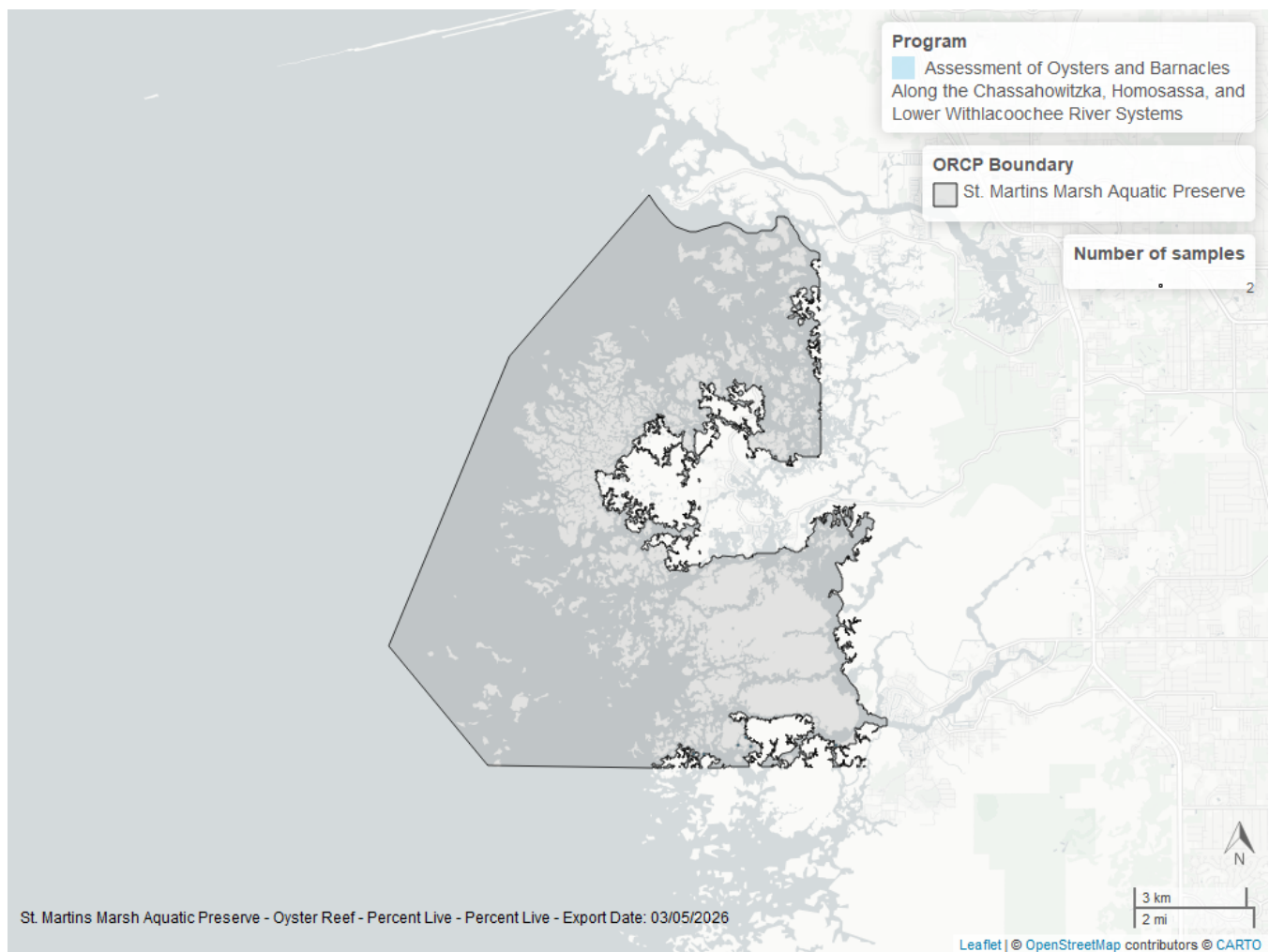


Figure 40: Map showing location of oyster percent live sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Shell Height

For natural reefs, there was insufficient data to calculate a trend for live oysters in either the 25-75mm or the ≥ 75 mm size class.

Natural

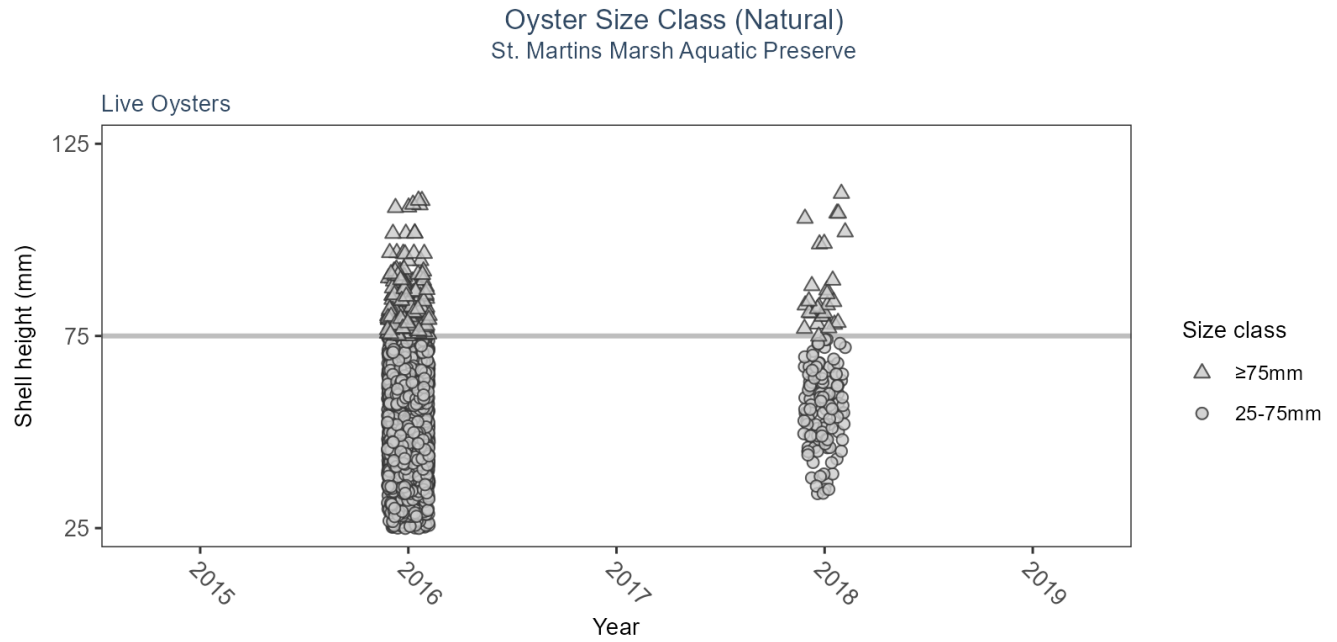


Table 39: Model results for Oyster Shell Height - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>SizeClass</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>No. of Samples</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Live Oysters	>75mm	Insufficient data	2016 - 2018	304	-	-	-
Natural	Live Oysters	25-75mm	Insufficient data	2016 - 2018	2679	-	-	-

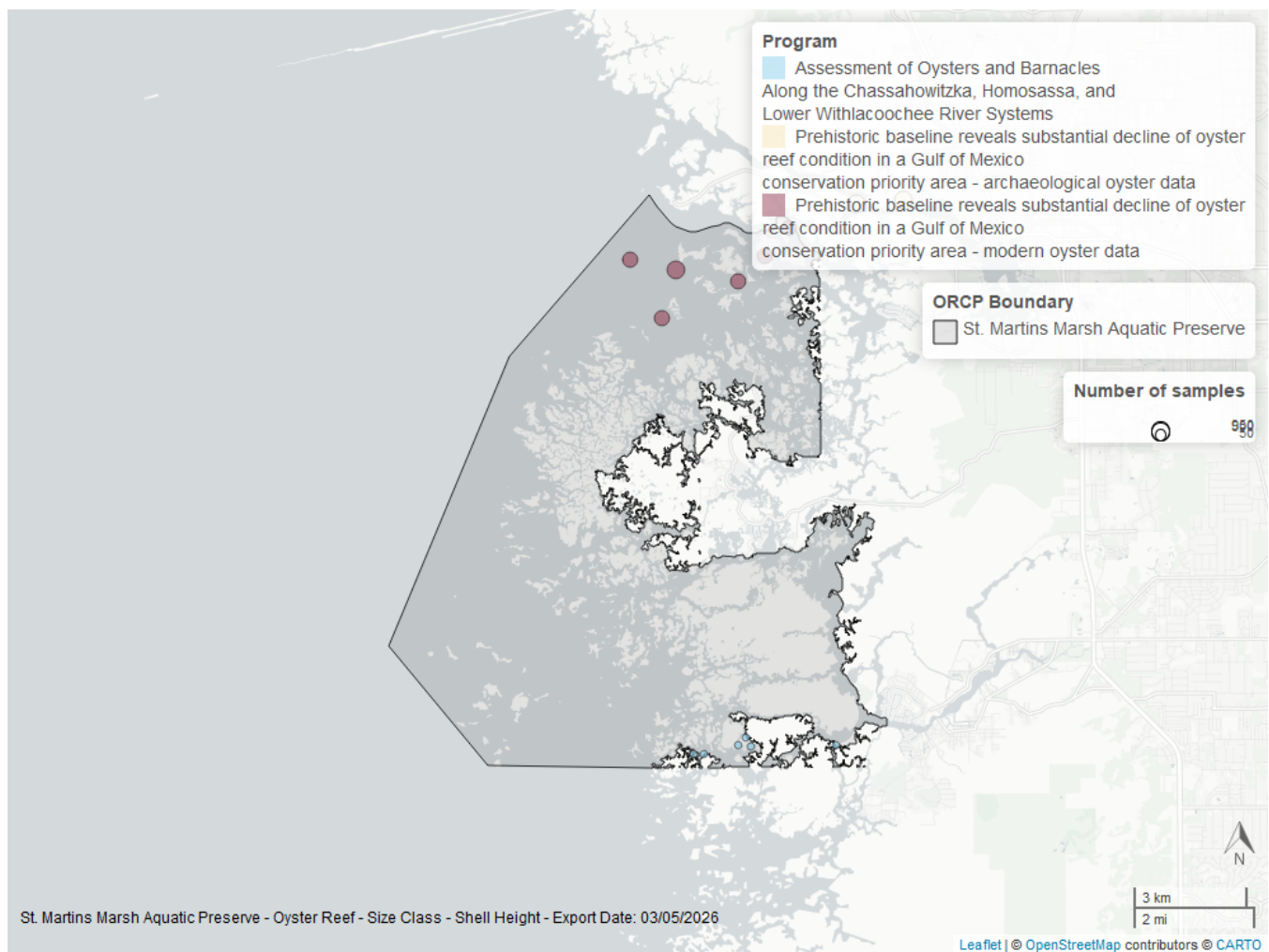


Figure 41: Map showing location of oyster shell height sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Acetabularia crenulata ¹	Digenea simplex ¹	Opsanus beta ²
Achirus lineatus ²	Drift algae ¹	Penaeus duorarum ²
Anadyomene stellata ¹	Echeneis neucratoides ²	Penicillus capitatus ¹
Anchoa hepsetus ²	Etropus crossotus ²	Penicillus dumetosus ¹
Anchoa mitchilli ²	Eucinostomus gula ²	Penicillus spp. ¹
Ariopsis felis ²	Eucinostomus harengulus ²	Prionotus scitulus ²
Avrainvillea levis ¹	Eucinostomus spp. ²	Prionotus tribulus ²
Bairdiella chrysoura ²	Gobiosoma bosc ²	Ruppia maritima ¹
Batophora oerstedii ¹	Gobiosoma robustum ²	Sargassum sp. ¹
Callinectes sapidus ²	Gobiosoma spp. ²	Sphoeroides nephelus ²
Caulerpa ashmeadii ¹	Halimeda incrassata ¹	Symphurus plagiusa ²
Caulerpa cupressoides ¹	Halodule wrightii ¹	Syngnathus louisianae ²
Caulerpa mexicana ¹	Halophila engelmannii ¹	Syngnathus scovelli ²
Caulerpa paspaloides ¹	Lagodon rhomboides ²	Synodus foetens ²
Caulerpa prolifera ¹	Leiostomus xanthurus ²	Syringodium filiforme ¹
Caulerpa sertularioides ¹	Lepisosteus osseus ²	Thalassia testudinum ¹
Chaetodipterus faber ²	Lucania parva ²	Trinectes maculatus ²
Chilomycterus schoepfii ²	Lutjanus griseus ²	Udotea flabellum ¹
Cliona caribbaea	Microgobius gulosus ²	Udotea spp. ¹
Codium isthmocladum ¹	No grass in quadrat ¹	Ulva sp. ¹
Cynoscion nebulosus ²	Ogcocephalus cubifrons ²	Unidentified species

1 - Submerged Aquatic Vegetation, 2 - Nekton

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